LEIYANG HEXI ANIMAL MANURE COMPOSTING PROJECT

Document Prepared by Climate Bridge (Shanghai) Ltd.

Tel: +86 21 6246 2036; Fax: +86 21 2301 9950

Project Title	**Leiyang Hexi Animal Manure Composting Project**
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Prepared By	Yujie Chen, Keming Lin, Yumei Peng, Mingying Fu Climate Bridge (Shanghai) Ltd.
Contact	Tel: +86 21 6246 2036; Email: projects@climatebridge.com Address: Block B, Level 24, Jiangong Mansion, No.33 Fushan Road, Pudong New Area, Shanghai, People's Republic of China 200120 Website: http://www.climatebridge.com

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Leiyang Hexi Animal Manure Composting Project (hereafter referred to as "the project") is a centralized manure treatment plant established in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City (county level), Hengyang City, Hunan Province, People's Republic of China (China), and is developed by Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd. (hereafter referred to as "the project proponent" or "the PP").

The project collects animal manure from 16 local swine farms in Leiyang City (county level) and removes solids through a solid-liquid separation system. The separated solids are further used to produce organic solid fertilizers through aerobic composting, effectively eliminating the potential for the manure solids to product methane emissions. The separated liquids are treated in water treatment systems to produce liquid fertilizers without being discharged into the fluvial system.

Prior to the implementation of the project, manure from swine barns was left to decay in uncovered anaerobic lagoons on each farm for uncontrolled anaerobic decay. After a minimum retention period of 6 months[1] to ensure full decomposition, the swine manure was ultimately utilized for land application in Leiyang City (county level). The lagoons received minimal maintenance, being rarely fully drained and cleaned, with the methane generated during decay released directly into the atmosphere without recovery.

The project has started operation since 28-Mar-2022. It introduces an aerobic animal manure composting system associated with solid separation techniques, avoids methane emissions through preventing manure solids from entering uncovered anaerobic lagoons, and further eliminates any potentials for methane emissions from manure solids through composting. The project is estimated to annually separate 52,500 tonnes of solids from swine manure, with an expected total organic fertilizer production of 30,000 tonnes.

Over the first 7-year project crediting period from 28-Mar-2022 to 27-Mar-2029, the project anticipates a total reduction of 401,149tCO_2e in GHG emissions, with an annual reduction of 57,307tCO_2e. During the 1st monitoring period from 28-Mar-2022 to 30-Sept-2023, the project reduced GHG emissions by 88,210tCO_2e.

Audit Type	Period	Program	VVB Name	Number of years

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[1] https://www.nahs.org.cn/gk/tz/202211/t20221103_413061.htm

Validation	-	Verified Carbon Standard	CTI Certification CO., LTD.	-
1st Verification	28-Mar-2022 to 30-Sept-2023	Verified Carbon Standard	CTI Certification CO., LTD.	1.5
Total	28-Mar-2022 to 30-Sept-2023	-	-	1.5

1.2 Sectoral Scope and Project Type

The project falls under Sectoral Scope 13: Waste handling and disposal.

The project is not an AFOLU project and is not a grouped project.

1.3 Project Eligibility

The scope of the VCS Program includes:

1) The seven Kyoto Protocol greenhouse gases: The Project is expected to avoid emissions of methane, which is a greenhouse gas designated by the Kyoto Protocol. Methane emissions from the anaerobic animal manure management system in the baseline scenario will be avoided or reduced in the project scenario.
2) Ozone-depleting substances: Not Applicable.
3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: Not Applicable.
4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The applied methodology AMS-III.Y (Version 04.0) of the project is a methodology approved under CDM Program, which is a VCS approved GHG program.
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not Applicable.

The project does not belong to projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.

Meanwhile, the project lies outside the scope of excluded project activities outlined in Table 1 of VCS Standard 4.5[2].

Therefore, the project is eligible under the scope of VCS program.

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[2] https://verra.org/wp-content/uploads/2023/08/VCS-Standard-v4.5-updated-11-Dec-2023.pdf

1.4 Project Design

The project has been designed to include a single location only.

The project is not being developed as a grouped project.

Eligibility Criteria

Not applicable.

1.5 Project Proponent

Organization name	Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd.
Contact person	Jun Li
Title	Project Manager
Address	Group 4, Dajin Village, Caizichi Street, Leiyang City (county level), Hengyang City, Hunan Province, People's Republic of China
Telephone	+86 2123019950
Email	3542346576@qq.com

1.6 Other Entities Involved in the Project

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultancy
Contact person	Zhiwen Gao
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, People's Republic of China
Telephone	+86 2162462036
Email	gao.zhiwen@climatebridge.com; projects@climatebridge.com

1.7 Ownership

The project owner is Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd., who has the legal right to control and operate the project activity. The ownership of the project is substantiated by the business license, the project approval, the Environmental Impact Assessment (EIA) approval, and the equipment purchasing contracts.

1.8 Project Start Date

According to the VCS standard v4.5[2], the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or CO_2 removals.

The start date of the project is on 28-Mar-2022, coinciding with the date when the solid separation system and the composting system were simultaneously and officially put into operation.

1.9 Project Crediting Period

The project adopts a 7-year crediting period that is twice renewable for a total of 21 years. The first crediting period of the project is from 28-Mar-2022 to 27-Mar-2029 (both days inclusive).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	X
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
28-Mar-2022 - 31-Dec-2022	43,805
01-Jan-2023 - 31-Dec-2023	57,307
01-Jan-2024 - 31-Dec-2024	57,307
01-Jan-2025 - 31-Dec-2025	57,307
01-Jan-2026 - 31-Dec-2026	57,307
01-Jan-2027 - 31-Dec-2027	57,307
01-Jan-2028 - 31-Dec-2028	57,307

01-Jan-2029 - 27-Mar-2029	13,502
Total estimated ERs	**401,149**
Total number of crediting years	**7**
Average annual ERs	**57,307**

1.11 Description of the Project Activity

The project serves to reduce GHG emissions in local swine farms across Leiyang City (county level) through the following procedures:

Manure collection

Upon receiving requests for manure treatment from farmers in the municipal area of Leiyang City (county level), the project owner deploys vacuum trucks to the designated farms to collect fresh manure slurry. The manure slurry is then transported in an airtight manner to the solid separation system located at the project site. The livestock farms involved, where the baseline adopts anaerobic processes without biogas recovery, are shown in Table 1-1 below:

Table 1-1 List of Livestock Farms

Location	Name	Year of Establishment	Business Scope
Sandu Town	Sandu Town Zhang Hong'an Pig Farm	2018	Swine farming
Donghu Town	Donghu Wei Town Dong Sanjun Pig Farm	2018	Swine farming
Donghu Town	Donghu Beilong Heping Pig Farm	2018	Swine farming
Gongping Town	Rixing Pig Farm	2017	Swine farming
Dayi Town	Hongsheng Pig Farm	2016	Swine farming
Yuqing Township	Yuqing Shixinyuan Pig Farm	2018	Swine farming
Donghu Town	Donghu Town Xinyuan Pig Farm	2016	Swine farming
Sandu Town	Sandu Town Yang Shaowu Pig Farm	2016	Swine farming
Xiaoshui Town	Xiaoshui Enrui Family Farm	2016	Swine farming
Gongping Town	Shunhe Family Farm	2018	Swine farming
Dayi Town	Dayi Town Hongli Ecological Pig Farm	2016	Swine farming
Dayi Town	Dayi Huang Xiaoying Pig Farm	2016	Swine farming
Dayi Town	Fengping Family Farm	2017	Swine farming

Dayi Town	Dayi Town Yushi Xingwang Livestock Farm	2016	Swine farming
Donghu Town	Donghu Wei Town Hu Xiaozhan Pig Farm	2018	Swine farming
Yuqing Township	Xu Xiaohong Pig Farm	2019	Swine farming

Solid separation system

The solid separation system is installed in the solid fertilizer workshop and is mainly comprised of two main components: a pre-treatment tank and mechanical solid-liquid separation equipment with screening and screw-pressing modules:

1) Vacuum trucks pump the manure slurry to an enclosed pre-treatment tank with an effective volume of 1500m^3, where auxiliary materials such as PH adjusters may be added to the manure slurry to meet the needs of solid separation. The pre-treated manure slurry usually remains a high moisture content (70% - 80%).
2) The pre-treated manure slurry is then discharged onto the top of the screening segment of the solid-liquid separator, which utilizes vibrating inclined screens that allow dilute manure slurry to flow through the screen openings. The flowing-down manure slurry from the screening segment is then fed in at one end of the screw-pressing segment. The screw compresses the manure slurry to further force water out, the separated solids usually have a moisture content at 60%.

No flocculants are needed before or during separation procedures.

The separated solids are aerobically composted on-site before any secondary application.

The separated liquids are diverted via pipes to the liquid fertilizer workshop, which is equipped with a water management system integrating both aerobic and anaerobic treatments. This system undergoes deammonification, denitrification, and concentration processes to produce high-efficiency liquid fertilizer. Methane generated during this process is not being recovered for utilization as the project system is not equipped with methane recovery system.

Composting system

The separated solids coming off the solid separation system immediately enter the composting system in the same workshop without the need for storage. Auxiliary materials such as fermentation strains and agricultural waste-based ingredients (plant ash, straw, rice hulls and etc.) may be added to the separated solids to meet the need of composting. The composting system consists of procedures including crushing, stirring and screening, all of which are carried out in an aerobic environment, effectively eliminating the potential for the manure solids to product methane emissions. The finished products are weighed, compressed and packed into baled bags by a baler installed in the solid fertilizer workshop, ready for sale.

The main equipment installed on-site are shown in Table 1-2 below:

Table 1-2 Main Equipment Installed

Equipment	Type	Number	Capacity	Technical Lifetime
Solid-liquid Separation Equipment	LX-950K-3Q-20hDAF-20	1	Screening: 0.75kw Screw-pressing: 3kw	15 yrs
Compost turner	RY-ZFP40B	1	50.7kw	15 yrs
Baler	RY-B50F	1	2.2kw	15 yrs

1.12 Project Location

The Project is located in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City (county level), Hengyang City, Hunan Province, China, and the project site is located at north latitude of 26°20’50.0” and east longitude of 112°48’35.8” specifically. The geographic location of the project is depicted in Figure 1-1. The satellite image was obtained in 2024 from Baidu Maps, as the most recent imagery from Google Satellite dates back to 2019, which is prior to the project implementation date.

Figure 1-1 Location of the project

1.13 Conditions Prior to Project Initiation

Prior to the project initiation, manure from swine barns was left to decay in uncovered anaerobic lagoons for uncontrolled anaerobic decay and was used for land application. The generated methane was released directly to the atmosphere without being recovered. The conditions prior to project implementation are the same as the baseline scenario, which is further demonstrated in Section 3.4 of this JPM.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with following relevant laws and regulations in China:

1) Environmental Protection Law of China[3]

2) Law of China on the Prevention and Control of Solid Waste Pollution[4]

3) Regulations on prevention and control of pollution from large scale livestock and poultry breeding[5]

4) Catalogue for the Guidance of Industrial Structure Adjustment (2019 revision)[6]

The project activity has been approved by the Development and Reform Bureau of Leiyang City (county level) initially on 16-Jul-2019 and revised on 22-Apr-2020 due to productivity changes of organic fertilizers. It has obtained the Environment Impact Assessment (EIA) approval from the Ecology and Environment Bureau of Hengyang City on 15-Jan-2021.

These two approvals well demonstrate that local government permits the construction of the project. Consequently, the project is compliance with laws, status and other regulatory frameworks.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only under VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading:

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[3] http://www.china-npa.org/uploads/1/file/public/201803/20180327160313_j8uwkhdxgn.pdf

[4] https://www.mee.gov.cn/ywgz/fgbz/fl/202004/t20200430_777580.shtml

[5] http://www.gov.cn/flfg/2013-11/26/content_2535095.htm

[6] http://www.gov.cn/xinwen/2019-11/06/content_5449193.htm

1) China has a national emissions trading scheme[7] that only targets high-emission industries such as thermal power generation, petrochemicals, chemicals, building materials, steel, non-ferrous metals, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO_2e/year. The project does not fall under the scope of this emissions trading scheme;

2) The project is not registered as a Chinese Certified Emission Reductions (CCER) project, and thus it does not qualify for emission reduction transactions within China's national emissions trading system. As a voluntary emission reduction initiative, any greenhouse gas emission reductions generated by the project will not be mandatory for compliance in China.

It is therefore ensured that the emission reductions during this monitoring period have not been and will not be accounted for in any emission trading programs or binding limitations.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received any other form of GHG-related credits, including renewable energy certificates.

1.16.3 Supply Chain (Scope 3) Emissions

According to the VCS standard v4.5[2], projects are not required to complete this section until the effective date of 01-Mar-2024. Therefore, this section is not applicable.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project collects animal manure from local swine farms in Leiyang City (county level) and removes solids through an animal manure composting system associated with solid separation techniques, avoids methane emissions through the separation of manure solids that are prevented from entering uncovered anaerobic lagoons. Specifically, the project contributes to sustainable development by achieving the following Sustainable Development Goals (SDGs):

8 DECENT WORK AND ECONOMIC GROWTH	SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all"[8]. China continuously improves the quality and efficiency of development. In-depth implementation of the innovation-driven development strategy, the rapid development of small and medium-sized enterprises. Adhering to the policy of giving priority to employment, the unemployment rate has remained

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[7] https://www.mee.gov.cn/xxgk2018/xxgk/xxgk06/202203/t20220315_971468.html

[8] https://sdgs.un.org/goals/goal8

	at a low level. By coordinating epidemic prevention and control and economic and social development, it has become the only major economy to achieve growth in 2020 and has made positive contributions to the recovery of the global economy. The project will provide new job opportunities and increase tax revenue, which will have a positive effect on the local economy. This contributes to one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value".
12 RESPONSIBLE CONSUMPTION AND PRODUCTION	SDG 12 "Ensure sustainable consumption and production patterns"[9]. China has made great efforts to develop a circular economy, publicizing, encouraging and promoting economical consumption patterns. To initially control the production and management of industrial solid waste and municipal waste. Through prevention, reduction, recycling and reuse, the generation of waste is greatly reduced, the recycling level of main waste is significantly improved, and the dependence on primary resources is reduced. The project introduces the aerobic composting system that is designed to compost animal manure from local swine farms and to produce organic fertilizers, ensuring sustainable consumption and production patterns. This contributes to one of China's stated sustainable development priorities "By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse".
13 CLIMATE ACTION	SDG 13 "Take urgent action to combat climate change and its impacts"[10]. In 2020, China's energy consumption per unit of GDP was reduced by 24.4% compared with 2012; carbon dioxide emissions per unit of GDP reduced by 18.8% compared with 2015 and 48.4% compared with 2005, all of which have already fulfilled China's commitment to the international community in 2020 ahead of schedule. The project will reduce CH_4 emissions from animal manure through removing the volatile solids and composting the separated solids. This contributes to one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in

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[9] https://sdgs.un.org/goals/goal12

[10] https://sdgs.un.org/goals/goal13

	agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas".

1.17.2 Sustainable Development Contributions Activity Monitoring

Throughout the 1st monitoring period, the project diligently adhered to the project description outlined in Section 1.11 of this JPM, achieving credible GHG emission reductions while contributing to SDG 8, 12 and 13 as demonstrated in Table 1-2.

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Table 1-2: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	8.5	The number of the job opportunities provided by the implementation of the project	Implemented activities to increase	Provided 18 long-term employment positions for local people during this monitoring period.	From 28-Mar-2022 onwards, the project has provided 18 long-term employment positions for local people.
2)	12.5	The amount of volatile solids (on a dry matter weight basis) contained in the animal manure waste treated by the project	Implemented activities to increase	Treated at least 17,355t[11] of volatile solids (on a dry matter weight basis) contained in the swine manure waste during this monitoring period.	From 28-Mar-2022 onwards, the project has treated at least 17,355t of volatile solids (on a dry matter weight basis) contained in the swine manure waste.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	Achieved a GHG emission reduction of 88,210CO_2e during this monitoring period.	From 28-Mar-2022 onwards, the project has achieved a GHG emission reduction of 88,210CO_2e.

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[11] Total volatile solids (t)=Dry matter (2022)×$VS_{ss,2022}$+Dry matter (2023)×$VS_{ss,2023}$=16,093.939t×0.55+15,184.739t×0.56=17,355t (rounded down to the nearest integer)

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

The EIA for the project was approved by the Ecology and Environment Bureau of Hengyang City on 15-Jan-2021 (Heng Huan Lei Ping [2021] No.15). This EIA report has assessed every possible aspect of environmental impact of the project and recommended corresponding mitigation measures, where applicable. No net harm on local environment and social community was detected for the project. The project also contributes to the sustainable development of local community as described in Section 1.17 of this JPM above.

2.2 Local Stakeholder Consultation

The PP hosted a LSC meeting in physical to provide direct interaction opportunities among stakeholders in May-2019. During the meeting, project information and corresponding environmental and social impacts of the project were communicated thoroughly to the local stakeholders. 30 copies of questionnaire were distributed to stakeholder representatives, taking account of different ages, genders and occupations as shown in Table 2-1 below.

Table 2-1 Structure of stakeholder survey for the project

Category		Count	Percentage
Number of stakeholders surveyed	Male	20	67%
	Female	10	33%
Age	<25	2	7%

	25-55	21	70%
	>55	7	23%
Education	Junior high school or below	22	73%
	Senior high school	6	20%
	College or above	2	7%
Occupation	Worker	8	27%
	Peasant	10	33%
	Management personnel	3	10%
	Civil servant	3	10%
	Unspecified	6	20%
Total		30	100%

The questionnaire listed a few multiple-choice questions regarding stakeholders' attitude towards the project implementation and its direct impacts. Table 2-2 below summarizes the questions and responses received. Most local stakeholder representatives believed that the project would improve local people's lives, promote local economic development, and have little impact on the environment. All stakeholder representatives were supportive to the project activities. The implementation of the project is considered beneficial.

Table 2-2 Summary of stakeholders' comments on the project

No.	Questions	Attitude or Opinion	Count	Percentage
1	Do you know about the project activity?	Very much	19	63%
		Heard of	8	27%
		Nothing	3	10%
2	Do you think the project will improve the natural environment?	Yes	27	90%
		No	0	0%
		Don't know	3	10%
3		Yes	18	60%

	Do you think the project will improve the financial situation?	No	6	20%
		Don’t know	6	20%
4	Do you think the project will improve the local employment situation and social community?	Yes	21	70%
		No	3	10%
		Don’t know	6	20%
5	What is the most probable environmental impact do you think the project will cause after implementation of the project?	None	29	97%
		Air pollution	0	0%
		Water pollution	0	0%
		Noise pollution	0	0%
		Harm to land	0	0%
		Don’t know	1	3%
6	What is your attitude to the project activity?	Support	30	100%
		Against	0	0%
		Indifferent	0	0%

The ongoing communications with local stakeholders have been proactively carried out at periodic intervals, with a commitment to keeping stakeholders informed about various aspects of project activities, including implementation, production, maintenance, monitoring, and the VCS (Verified Carbon Standard) verification process. Comments and suggestions from any of these entities are collected and considered by the PP. During the 1st monitoring period of the project, the PP conducted the questionnaire survey again to gather feedback from local stakeholder representatives in Nov-2022. The survey received a 100% response rate. No negative comments were reported.

The PP has also implemented a comprehensive approach to engage with local stakeholders by introducing the project information and establishing a continuous inputs/grievance mechanism. Stakeholders are able to report grievances or provide comments by utilizing the grievance book available at the project office on-site, or by reaching out directly via phone or email at any time. The received inputs and grievances are recorded, responded, archived, and a designated staff member, appointed by the project proponent, oversees the collection and management of inputs/grievance form local stakeholders, and the relevant solution process is as follows:

1) Firstly, once getting the complaints or grievances, project proponent shall contact and discuss with relevant community or other stakeholders within 3 days;

2) Then the specific staff of project proponent should propose a solution and mediation which is performed by relevant government agency within a week based on all collected information;

3) Thirdly, the project proponent will address their complaints or grievances by relevant legal method such as arbitration and courts if necessary. Finally, the entire complaints or grievances redress process shall be dealt within 30 days.

The local government agencies and competent authorities also conducted spot checks on the implementation of the project at periodic intervals and provided suggestions on the involved rectification problems if needed.

During the 1st monitoring period, no inputs/grievances received through all the methods.

2.3 Environmental Impact

The EIA report of the project was approved by Ecology and Environment Bureau of Hengyang City on 15-Jan-2021 (Heng Huan Lei Ping [2021] No.15). The EIA report comprehensively evaluates every possible aspect of the project's environmental impact and, where applicable, proposes corresponding mitigation measures, as outlined below:

Wastewater

The production wastewater of this project includes rainwater, surface cleaning wastewater, and production wastewater (wastewater from equipment and vehicle cleaning). They are collected and enter the pre-treatment tank without being discharged into external fluvial systems. The transportation of manure utilizes specialized, enclosed vehicles, and storage and transportation procedures are strictly established to ensure cleanliness on both the interior and exterior grounds of the project site. The solid fertilizer workshop is designated as a general impermeable area.

Air pollution

The main pollutants include odorous gases (NH_3, H_2S, etc.) and dust generated during the composting process in the solid fertilizer workshop. The odorous gases are purified through a biofilter tower, meeting the second-level standard of the “Odor Pollutants Emission Standard” (GB14554-93), and are discharged through a 15m exhaust pipe. The exhaust gases from the crushing and screening processes of composting undergo dust collection and are treated by a bag filter to meet the second-level standard of the “Comprehensive Emission Standard for Air Pollutants” (GB16297-1996). These gases are discharged through a 15m exhaust pipe. The workshop is fully enclosed, equipped with ventilation facilities. The project should establish a sanitary protection distance of 100m around the solid fertilizer workshop. Greenspace with trees and shrubs should be settled surrounding the workshop.

Noise

The primary sources of noise include operational sounds from equipment, such as grinding and cutting pumps, crushers, and screening machines, as well as transportation noise from vehicles. Measures such as reinforcement and vibration reduction, installation of soundproofing devices, sealing, and rational scheduling of production time should be implemented. The noise levels must comply with the requirements of Class 2 standards outlined in the "Industrial Enterprise Boundary Environmental Noise Emission Standards" (GB12348-2008). The transportation volume and time period are strictly regulated, no transport vehicle is allowed to enter the project site during nighttime (10p.m. – 6a.m.).

Solid waste

The dust collected by the bag filter is reintroduced into the production of solid fertilizer. There will also be a certain amount of waste packaging generated, which are recycled by waste recycling stations.

To conclude, the project is not anticipated to have significant environmental impacts during the operation period. During the 1st monitoring period, mitigation measures were properly implemented and no significant impact on the surrounding environment was found.

2.4 Public Comments

This project has been listed for public comments on the verra website from 22-Nov-2023 to 22-Dec-2023 and no comments were received.

2.5 AFOLU-Specific Safeguards

Not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project follows the VSC Standard version 4.5 and uses the following methodologies and tools.

- CDM Small-scale Methodology AMS-III.Y: “Methane avoidance through separation of solids from wastewater or manure treatment systems”, version 04.0

  https://cdm.unfccc.int/methodologies/DB/IR1ULTHWQKPQO992O3UJTTLELME23L

- CDM Small-scale Methodology AMS-III.F: “Avoidance of methane emissions through composting”, version 12.0

  https://cdm.unfccc.int/methodologies/DB/NZ83KB7YHBIA7HL2U1PCNAOCHPUQYX

- CDM Small-scale Methodology AMS-I.D: “Grid connected renewable electricity generation”, version 18.0

https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK

- CDM Methodological Tool 05: "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation", version 03.0

  https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf/history_view

- CDM Methodological Tool 13: "Project and leakage emissions from composting", version 02.0

  https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-13-v2.pdf/history_view

- CDM Methodological Tool 21: "Demonstration of additionality of small-scale project activities", version 13.1

  https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf/history_view

3.2 Applicability of Methodology

The project meets the applicability criteria outlined in the methodology and associated tools in the following ways.

- CDM Small-scale Methodology AMS-III.Y: "Methane avoidance through separation of solids from wastewater or manure treatment systems", version 04.0:

Para.	Criteria	Justification for the Project Activity
2	This methodology comprises technologies and measures that avoid or reduce methane production from anaerobic wastewater treatment systems and anaerobic manure management systems, through removal of (volatile) solids from the wastewater or manure slurry stream. The separated solids shall be further treated, used or disposed in a manner resulting in lower methane emissions.	Applicable. The project collects animal manure from local swine farms in Leiyang City (county level) and removes solids through a solid-liquid separation system to avoid methane emissions generated in open anaerobic lagoons. The separated solids are further used to produce organic solid fertilizers through aerobic composting onsite before any secondary use. The project is estimated to annually separate 52,500 tonnes of solids from swine manure, with an expected total organic fertilizer production of 30,000 tonnes.

3	The project activity does not recover and combust biogas i.e. the baseline wastewater or manure treatment plant as well as the project system are not equipped with methane recovery system.	Applicable. Neither the baseline nor project activity are equipped with methane recovery system. Prior to the implementation of the project, manure from swine barns was left to decay in uncovered anaerobic lagoons on each farm for uncontrolled anaerobic decay. The methane produced during decay is released directly into the atmosphere without undergoing any recovery process. The project activity introduces an aerobic animal manure composting system associated with solid separation techniques, yet it does not include a methane recovery system.
4	The technology for solids separation shall be one of the below or a combinations thereof so as to achieve a minimum dry matter content of separated solids larger than 20% a) A pre-separation phase of chemical treatment by mixing flocculants with the wastewater, adopted to improve the efficiency of the subsequent mechanical solid-liquid separation process; b) Mechanical solid/liquid separation technologies (e.g. stationary, vibrating or rotating screens, centrifuges, hydrocyclones, press systems/screws), operated in-line with the inflowing freshly generated wastewater or slurry manure stream so as to avoid stagnation;	Applicable. The project involves the implementation of b) mechanical solid/liquid separation technologies. The equipment for this project complies with the requirements of industry standard JB/T 13756-2019 General Technical Conditions for Solid-Liquid Separation of Livestock and Poultry Manure. According to the standard, “when the separator machine separates solid materials using the filter specified in the product manual under normal temperature and pressure, the overall performance of the machine should meet the following criteria: for separators used for flushing (bubbling) manure, the moisture content of the separated solid material should be ≤75%. This ensures that the minimum dry matter content of the

	c) Thermal treatment technologies that evaporate water content from the waste stream, either releasing vapour to the atmosphere or condensing it into a liquid fraction (condensate). Examples include evaporation and spray drying technologies.	separated solids is greater than the requirement of 20%. Additionally, based on the Solid-liquid separation records provided by the PP, the moisture content of the separated manure after solid-liquid separation is around 60%, which further confirmed that the dry matter content of separated solids could remain higher than 20% throughout until its final use as fertilizers.
5	The dry matter content of separated solids shall remain higher than 20% throughout until its final disposal, destruction or use (e.g. spreading on the soil). The total time interval for the separation process until 20% dry matter content is achieved shall be less than 24 hours.	Applicable. The dry matter content of separated solids exceeds 20%, and the separation process for each batch of the manure slurry takes less than 24 hours.
6	Separation of solids using gravity (settling tanks/basins, ponds, or geotextile containers/bags) is not included in this methodology.	Applicable. The project does not involve the implementation of gravity-drive separation technologies.
7	In case of animal manure management systems the following conditions apply: a) Animals shall be managed in confined conditions; b) If organic bedding material is used in the animal barns or added to the manure stream, separated solids shall not be reused in the barns; c) If the baseline manure slurry was treated in an anaerobic lagoon or another liquid treatment system, the outflow liquid from the lagoon was recycled as flush water or used to irrigate fields; however it was not discharged into river/lake/sea. In the	Applicable. The project involves animal manure management systems with the following conditions applied: a) The project serves for local farms that have a high productivity, where animals are managed in confined conditions; b) The project utilizes the separated solids to produce organic fertilizers through composting and will not reuse separated solids in the barns; c) The baseline manure slurry was treated in uncovered anaerobic

	latter case, i.e. effluent discharge into river/lake/sea, the system is considered as a wastewater treatment system and not a manure management system; d) A minimum interval of six months was observed between each removal of the solids accumulated in the lagoon.	lagoons and the outflow liquid from the lagoon was used to irrigate fields; d) A minimum interval of six months[1] was observed between each removal of the solids accumulated in the lagoons.
8	In case of wastewater treatment systems the following conditions apply: The baseline treatment systems do not include a fine solids separation process (i.e. grading smaller than 10 mm aperture, primary settlers, mechanical separation, etc.); In case the baseline treatment system was an anaerobic lagoon or a liquid system, a minimum interval of 30 days was observed between each removal of the solids accumulated in the lagoon.	Irrelevant. There is no water treatment system involved in this project. While the baseline manure slurry was treated in uncovered anaerobic lagoons, the outflow liquid from the lagoon was recycled and used for land application. As it was not discharged into river/lake/sea, the system is considered as a manure management system and not a wastewater treatment system.
9	This methodology is not applicable when the project treats solids removed from an existing lagoon, or sludge originated from settlers or from any other biologically active treatment device of the baseline animal manure management/wastewater treatment system.	Irrelevant. The project does not treat solids removed from an existing lagoon, or sludge originated from settlers or from any other biologically active treatment device.
10	The separated solids shall be further treated and methane emissions resulting from further treatment, storage, use or disposal shall be considered. If the solids are combusted for thermal or heat generation, that component of the project activity can use a corresponding methodology under Type I. If the solids are mechanically/thermally treated to produce refuse-derived fuel (RDF) or stabilized biomass (SB) the relevant provisions in	Applicable. The separated solids are further composted to product organic solid fertilizers, and the resulting methane emissions are accounted for as part of the project emissions.

	AMS-III.E shall be followed. If the solids are used as animal feeds (e.g. feed to cows, pigs), emissions from enteric fermentation and emissions from the manure may be neglected.	
11	The liquid fraction from the project solid separation system shall be treated either in the baseline treatment facility or in a treatment system with a lower methane conversion factor (MCF) than the baseline system.	Applicable. The liquid fraction from the project solid separation system is diverted to produce liquid fertilizers in a water management system that combines aerobic and low-leakage anaerobic treatments, which has a lower MCF than the baseline uncovered anaerobic lagoons referring to Table 10.17 from 2019 Refinement to the 2006 IPCC Guidelines.
12	This methodology applies to situations where the baseline treatment systems have been operational at least three years before the start date of the project activity. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in an efficiency improvement or capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the requirements in the “General Guidelines to SSC CDM methodologies” concerning these topics. In addition, the requirements for demonstration of the remaining lifetime of the equipment	Applicable. The project collects animal manure from existing local swine farms in Leiyang City (county level) and removes solids through a solid-liquid separation system, while the baseline treatment system, in which the manure solids being left in uncovered anaerobic lagoons, was a very traditional and long-standing common practice among local swine farms. Based on a review of publicly available information provided by the National Enterprise Credit Information Publicity System (developed by an officially registered enterprise credit agency, Qichacha[12]), it was verified all the entities operating all the involved swine farms have been

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[12] https://www.qcc.com/

	replaced as described in the general guidelines shall be followed.	operational for at least three years prior to the implementation date of the project on 28-Mar-2022.
13	In case flocculants is used in the project activities, project emissions and leakage from use of flocculants should be taken into account.	Irrelevant. The project does not use flocculants in the project activity.
14	Measures are limited to those that result in emission reductions of less than or equal to 60 Kt CO_2 equivalent annually.	Applicable. The annual emission reductions of the project do not exceed 60 Kt CO_2.

- CDM Small-scale Methodology AMS-III.F: “Avoidance of methane emissions through composting”, version 12.0: The primary methodology AMS-III.Y directs the project to use this methodology to quantify project emissions from composting of solid separated.
- CDM Small-scale Methodology AMS-I.D: “Grid connected renewable electricity generation”, version 18.0: The primary methodology AMS-III.Y directs the project to use this methodology to quantify project emissions from energy consumption of solid separation.
- CDM Methodological Tool 13: “Project and leakage emissions from composting”, version 02.0. The methodology AMS-III.F directs the project to use this tool to quantify project emissions from composting of solid separated.

Para.	Criteria	Justification for the Project Activity
2	Typical applications of the tool include projects composting municipal solid wastes, agricultural wastes and digestate.	Applicable. The project activity introduces a newly built aerobic animal manure composting system to treat the animal manure from local swine farms in Leiyang City (county level).
3	The following sources of project emissions are accounted for in this tool: a) CH_4 and N_2O emission from composting; b) CO_2 emissions from consumption of fossil fuels and electricity associated with composting; and	Applicable. a) CH_4 and N_2O emissions from composting are taken into consideration; b) CO_2 emissions from consumption of fossil fuels and electricity

	c) CH_4 emissions from run-off wastewater associated with co-composting.	associated with composting are taken into consideration; c) CH_4 emissions from run-off wastewater associated with co-composting do not exit.
4	The following source of leakage emissions is accounted for in this tool: a) CH_4 emissions from the anaerobic decay of the residual organic content of compost disposed of in a landfill or subjected to anaerobic storage.	Irrelevant. Any residual organic content of compost is placed on-site in aerobic condition. It is neither disposed of in a landfill nor subjected to anaerobic storage.
5	Transport emissions are not accounted for in this tool because it is assumed that similar transportation activities would occur in the baseline.	Applicable. Transport emissions are not taken into consideration.

- CDM Methodological Tool 05: “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0. The methodology AMS-III.F directs the project to use this tool to quantify project emissions from electricity consumption by composting of solid separated.

Para.	Criteria	Justification for the Project Activity
5	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer.	Applicable. The project calculates emissions from electricity consumption. Scenario A is applicable as the project consumes electricity from the local power grid connected to the Central China Power Grid (CCPG).

	b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
6	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: a) Scenario I: Electricity is supplied to the grid; b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities	Irrelevant. The project does not involve electricity generation in the project scenario.
7	This tool is not applicable in cases where captive renewable power generation	Irrelevant.

	technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO_2 emissions.	The project does not involve installation of captive renewable power generation technologies.

- CDM Methodological Tool 21: "Demonstration of additionality of small-scale project activities", version 13.1:

Para.	Criteria	Justification for the Project Activity
4	The use of the methodological tool "Demonstration of additionality of small-scale project activities" is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	Irrelevant. The PP will not propose any new methodologies and will not propose any alternative methods to demonstrate additionality.
5	Project participants and coordinating/managing entities may also apply "TOOL19: Demonstration of additionality of microscale project activities" as applicable.	Irrelevant. The project is a small-scale project as the annual emission reductions of the project does not exceed 60 Kt CO_2 threshold designated by CDM.

3.3 Project Boundary

As per para.20 of AMS-III.Y, the project boundary applicable to the proposed project activity is the physical, geographical site of:

1) Previously uncovered anaerobic lagoons on local swine farms, where, prior to the project's initiation, swine manure was collected and treated, resulting in methane emissions, and are no longer utilized for manure treatment after the project's initiation;
2) The project site where the treatment of animal waste through solids separation takes place;
3) The project site where the composting of the separated solids takes place;
4) The site where land application of the separated solids takes place;
5) The transportation itineraries for matters within the project boundary.

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from the waste treatment process	CO_2	No	Excluded for simplification. This emission source is assumed to be very small
		CH_4	Yes	The major emission source
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small
		Other	No	Excluded for simplification. This emission source is assumed to be very small
Project	Project emissions from storage, use, destruction or disposal of solids separated	CO_2	Yes	Involves emission from composting, which may be an important emission source
		CH_4	Yes	Involves emission from composting, which may be an important emission source
		N_2O	Yes	Involves emission from composting, which may be an important emission source
		Other	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from electricity and fossil fuel use by the solid separation technology	CO_2	Yes	May be an important emission source
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small
		Other	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from the combustion of flocculants used in the pre-separation phase	CO_2	No	Does not exist
		CH_4	No	Does not exist
		N_2O	No	Does not exist
		Other	No	Does not exist
		CO_2	Yes	May be an important emission source

Source		Gas	Included?	Justification/Explanation
	Incremental emissions due to increased transportation for matters	CH_4	No	Excluded according to AMS-III.Y
		N_2O	No	Excluded according to AMS-III.Y
		Other	No	Excluded according to AMS-III.Y

In addition, Figure 3-1 below further depicts the project boundary and activities.

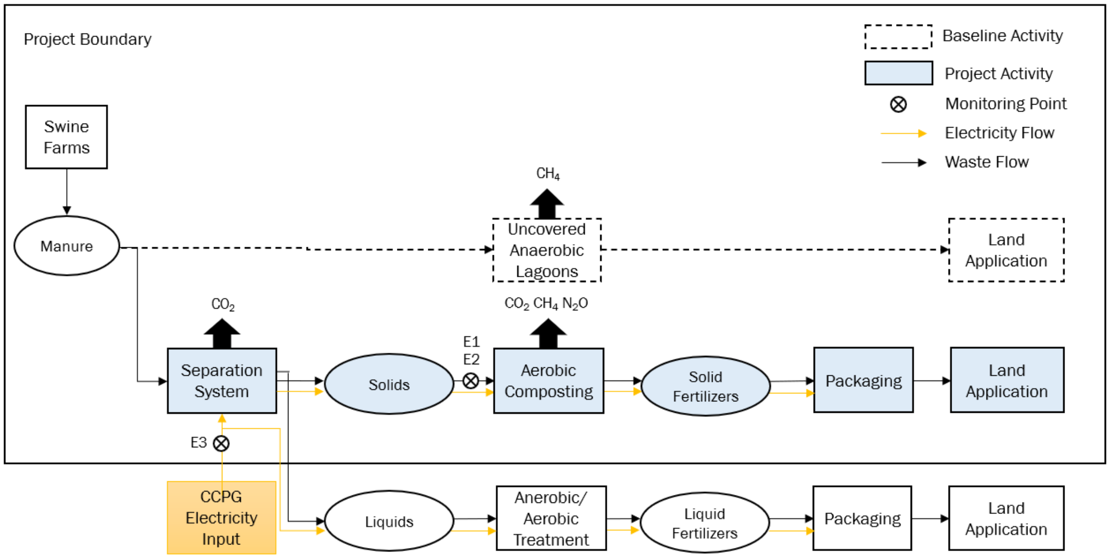

Figure 3-1 Project Boundary

3.4 Baseline Scenario

In the baseline scenario of this project, manure from swine barns was left to decay in uncovered anaerobic lagoons on each farm for uncontrolled anaerobic decay. After a minimum retention period of 6 months[1] to ensure full decomposition, the swine manure was ultimately utilized for land application in Leiyang City (county level). The lagoons received minimal maintenance, being rarely fully drained and cleaned, with the methane generated during decay released directly into the atmosphere without recovery.

3.5 Additionality

The project activity, which introduces an aerobic animal manure composting system associated with solid separation techniques to avoid methane emissions through the separation of manure solids, is not mandated by any of the identified regulatory frameworks listed in Section 1.14, fulfilling requirements of regulatory surplus.

As per para.17 of AMS-III.Y, the PP shall take into account the general guidance to the small-scale (SSC) CDM methodologies and the Tool for demonstration of additionality of SSC project activities. The project therefore applies Tool 21 to demonstrate additionality due to the project scale, which does not exceed the small-scale CDM threshold of 60 Kt CO_2.

Tool 21 requires the PP to demonstrate at least one obstacle included in the tool to show that the project activity would not have occurred anyway. The investment barrier was identified as the most relevant barrier.

The PP conducted a benchmark analysis based on the internal rate of return (IRR) of project cash flows to assess the financial position of the proposed project activities in the absence of the revenue from carbon credit. The revenue source of the project is the production and sales of organic fertilizers, which is classified into the fertilizer industry according to the "Industrial Classification for National Economic Activities" (GB/T 4754-2017)[13]. Detailed procedures of the benchmark analysis are demonstrated below.

In April 2019, during the preparation of the initial project design report, the PP carefully considered pursuing emission reduction benefits to enhance the project's financial status and decided to develop this project as a voluntary emission reduction project under VCS mechanism.

IRR calculation

Basic parameters applied for IRR calculation are listed in Table 3-1:

Table 3-1 Basic Parameter of IRR Calculation

Parameter	Value	Source
Annual organic fertilizer production	30,000 t/yr	Feasibility Study Report
Price of organic fertilizer	380 CNY/t	Feasibility Study Report
Static investment	28,000,000 CNY	Feasibility Study Report
Annual O&M cost	8,362,277 CNY/yr	Feasibility Study Report
Maintenance cost	549,580 CNY/y	Feasibility Study Report
Salary and welfare	1,944,000 CNY/y	Feasibility Study Report
Insurance	68,697 CNY/y	Feasibility Study Report
Material cost	4,000,000 CNY/y	Feasibility Study Report

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[13] https://max.book118.com/html/2021/1017/8140033046004021.shtm

Other cost	1,800,000 CNY/y	Feasibility Study Report
Operation Period	15 yrs	Feasibility Study Report
Construction Period	1 yr	Feasibility Study Report
Rate of residual value	5%	Feasibility Study Report
VAT tax rate for organic fertilizers sales	9%	Feasibility Study Report
Tax refund for organic fertilizer sales	100%[14]	Feasibility Study Report
VAT of materials purchasing	13%	Feasibility Study Report
Urban maintenance and construction tax	1%	Feasibility Study Report
Surtax for education	5%	Feasibility Study Report
Income tax	25%	Feasibility Study Report
Emission reductions	57,307 tCO_2e/yr	ER calculation
Price of VCUs	25 CNY/tCO_2e	Expected

The calculation result of the project's IRR is 5.42%.

1) Comparing project's IRR with post-tax equity IRR benchmark for the fertilizer industry

According to the "Table of the Financial Benchmark Rate of Return for Project Constructions" issued by the National Development and Reform Commission of China, the post-tax equity IRR benchmark for the fertilizer industry is 10%[15]. The project's IRR of 5.42% is much lower than the post-tax equity IRR, suggesting that the proposed project is financially unacceptable due to such a low profitability and thus investment barrier occurs.

Table 3-2 Comparison of Post-tax equity IRR with and without income from VCUs

Without income from VCUs	Benchmark	With income from VCUs
5.42%	10%	11.12%

2) Sensitivity analysis

The sensitivity analysis is conducted to confirm whether, under reasonable deviations in the four financial indicators stated below, the project's IRR would not exceed the post-tax equity IRR

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[14] http://www.mof.gov.cn/zhengwuxinxi/zhengcefabu/2008zcfb/200806/t20080603_44622.htm

[15] https://www.ndrc.gov.cn/fggz/gdzctz/tzfg/201907/t20190729_1197578.html?code=&state=123

without receiving revenue from carbon credit, and thus would still remain financially unattractive. The four financial indicators that significantly affect the project's IRR are:

a) Static investment

b) Annual organic fertilizer production

c) Annual O&M cost

d) Price of organic fertilizer

The impacts of the above indicators on the project's IRR were analyzed. The results of this sensitivity analysis are shown in Table 3-3.

Table 3-3 Sensitivity analysis of the project's IRR to different financial indicators

Hypothetical change	–10%	-5%	0	+5%	+10%
Static investment	7.01%	6.18%	5.42%	4.71%	4.06%
Annual organic fertilizer production	0.44%	3.00%	5.42%	7.73%	9.97%
Annual O&M cost	8.80%	7.13%	5.42%	3.65%	1.81%
Price of organic fertilizer	0.44%	3.00%	5.42%	7.73%	9.97%

The post-tax equity project IRR could reach the benchmark of 10% if one of the following conditions are achieved:

a) The static investment decreases by 24.98%, equivalent to 6,993,345 CNY. However, this is hardly achievable, as such a significant reduction would likely require elimination of equipment quantity and quality, which may lead to a decline in productivity, ultimately impacting profit margins. The actual static investment of 2022 (Mar-2022 to Feb-2023) was 27,986,200 CNY/yr, which slightly lower than the expected production of 28,000,000 CNY;

b) The annual organic fertilizer production increases by 10.04%, equivalent to 3,012t. However, this is hardly achievable. With the fixed capacity of the composting system, the output of organic fertilizer is consequently expected to remain stable as well. The actual fertilizer production of 2022 (Mar-2022 to Feb-2023) was 26,925 t/yr, which lower than the expected production of 30,000 t/yr;

c) The annual O&M cost decreases by 13.63%, equivalent to 1,140,129 CNY. However, this is hardly achievable, as there has been a notable upward trend in material cost[16]

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[16] https://www.stats.gov.cn/sj/ndsj/2023/indexch.htm

and labour cost[17] during the construction and implementation period. The actual annual O&M cost of the project in 2022 (Mar-2022 to Feb-2023) was 8,352,173 CNY/yr, which slightly lower than the expected cost of 8,362,277 CNY/yr;

d) The price of organic fertilizer increases by 10.04%, equivalent to 38 CNY. However, this is hardly achievable, as the organic fertilizers are sold to local customers under signed fixed-price contracts with prices that reflect a thorough consideration of costs, while any attempt to increase prices may result in reduced demand as customers may seek more affordable alternatives. The actual selling price was 400 CNY per tonne, yielding an IRR of 7.85%, which still fell below the benchmark.

3) Conclusion

As evidenced by the financial statistics presented herein, the project faces an investment barrier and is therefore considered additional.

3.6 Methodology Deviations

Not applicable.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The project started construction on 28-Aug-2019 and has been officially in operation since 28-Mar-2022. The COVID-19 pandemic considerably delayed the construction schedule; hence the completion of the project construction exceeded the initial anticipated timeframe.

This is the 1st monitoring period of this project spanning from 28-Mar-2022 to 30-Sept-2023 with both days included. Throughout this monitoring period, the project diligently adhered to the project description outlined in Section 1.11 of this JPM with no significant alterations to the project design, and it experienced no malfunctions or technical failures. No events that may impact the GHG emission reductions or removals and monitoring occurred during this monitoring period. The key system involved in the project are as follows:

Manure collection

Upon receiving requests for manure treatment from farmers in the municipal area of Leiyang City (county level), the project owner deployed vacuum trucks to the designated farms to collect fresh manure slurry. The manure slurry was then transported in an airtight manner to the solid separation system located at the project site.

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[17] https://www.stats.gov.cn/sj/ndsj/2023/indexch.htm

Solid separation system

The solid separation system has been installed in the solid fertilizer workshop and is mainly comprised of two main components: a pre-treatment tank and mechanical solid-liquid separation equipment with screening and screw-pressing modules. During this monitoring period, the solid separation system has been operated as follows:

1) Vacuum trucks pumped the manure slurry to an enclosed pre-treatment tank with an effective volume of 1500m^3, where auxiliary materials such as PH adjusters were added to the manure slurry to meet the needs of solid separation. The pre-treated manure slurry had a high moisture content (70% - 80%).
2) The pre-treated manure slurry was then discharged onto the top of the screening segment of the solid-liquid separator, which utilized vibrating inclined screens that allowed dilute manure slurry to flow through the screen openings. The flowing-down manure slurry from the screening segment was then fed in at one end of the screw-pressing segment. The screw compressed the manure slurry to further force water out, the separated solids consistently had a moisture content at around 60%.

No flocculants were needed before or during separation procedures. The separated solids were aerobically composted on-site before any secondary application, while the separated liquids were diverted via pipes to the liquid fertilizer workshop, where a water management system that combined aerobic and anaerobic treatments were installed to produce organic liquid fertilizers.

Composting system

The separated solids coming off the solid separation system immediately entered the composting system in the same workshop without the need for storage. Auxiliary materials such as fermentation strains and agricultural waste-based ingredients (plant ash, straw, rich hull and etc.) were added to the separated solids to meet the need of composting. The composting system consisted of procedures including crushing, stirring and screening, all of which were carried out in an aerobic environment, effectively eliminating the potential for the manure solids to product methane emissions. The finished products were weighed, compressed and packed into baled bags by a baler installed in the solid fertilizer workshop, ready for sale.

The main equipment installed on-site are shown in Table 4-1 below:

Table 4-1 Main Equipment Installed

Equipment	Type	Number	Capacity	Technical Lifetime
Solid-liquid Separation Equipment	LX-950K-3Q-20hDAF-20	1	Screening: 0.75kw Screw-pressing: 3kw	15 yrs
Compost turner	RY-ZFP40B	1	50.7kw	15 yrs

Baler	RY-B50F	1	2.2kw	15 yrs

The project separated 78,555 tonnes of solids from swine manure over the 1st monitoring period, reducing GHG emissions by 88,210tCO$_2$e.

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The barns within the project boundary do not use any organic bedding materials, therefore baseline emissions are calculated based on the total mass of the volatile solids separated as per para.23 of AMS-III.Y:

$$\mathrm{BE_y} = \left(\mathrm{B_{o,w,y}} \times \mathrm{M_{ss,y}} \times \mathrm{VS_{ss,y}} \times \mathrm{UF_b} \times \mathrm{GWP_{CH4}} \times \mathrm{D_{CH4}}/1000\right) \times \sum_{\mathrm{i}} \left(\mathrm{MS_{Bl,i}} \times \mathrm{MCF_{b,i}}\right) \quad (1)$$

Where:

$\mathrm{BE_y}$ = Baseline emissions in year y (tCO$_2$e)

$\mathrm{B_{o,w,y}}$ = Weighted methane-producing potential of the volatile solids separated by the project in year y (m^3 CH$_4$ per kg of VS). The project only involves manure from swine ($\mathrm{B_{o,w,y}} = \mathrm{B_{o,swine,y}}$).

$\mathrm{M_{ss,y}}$ = Mass (dry matter basis) of total separated solids in year y (kg)

$\mathrm{VS_{ss,y}}$ = Volatile solids content of the separated solids in year y on a dry matter basis (kg/kg)

$\mathrm{UF_b}$ = Model correction factor to account for model uncertainties (0.94)[18]

$\mathrm{GWP_{CH4}}$ = Global Warming Potential of methane applicable to the crediting period (tCO$_2$e/tCH$_4$)

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[18] Reference: FCCC/SBSTA/2003/10/Add.2, page 25

D_{CH4} = Conversion factor of m^3 CH_4 to kilograms (0.67 kg per m^3 at 20°C and 1 atm pressure)

i = Index for baseline anaerobic manure management system

$MS_{Bl,i}$ = Fraction of manure handled in the baseline anaerobic manure management system i (fraction, mass basis), based on historical information from previous year(s) before project. If historical information is not available, the capacity (t/day) of each baseline management system shall be verified before the project start. During the project, it would be assumed that the management systems with lower MCFs would be used up to their full capacity, and only thereafter the systems with larger MCF would be used. The value of $MS_{Bl,i}$ is "1.0" if only one type of treatment system was used in the baseline

$MCF_{b,i}$ = Methane conversion factor for the baseline anaerobic manure management system i as per IPCC Guidelines for National Greenhouse Gas Inventories

5.2 Project Emissions

Project emissions are quantified according to AMS-III.Y and AMS-III.F.

According to para.29 of AMS-III.Y, project activity emissions consist of:

1) Any methane emissions from storage, use, disposal or destruction of solids separated ($PE_{y,ss}$);
2) Emissions from electricity and fossil fuel use by the solid separation technology as per the methods of AMS-I.D ($PE_{y,power}$);
3) Emissions from the combustion of flocculants used in the pre-separation phase ($PE_{y,floc,combustion}$);
4) Incremental CO_2 emissions due to increased transportation ($PE_{y,transp}$):
    a) Transportation of solids to sites where it will be treated further or gainfully used (within the project boundary);
    b) Transportation of solids from and to treatment facilities to storage sites (within the project boundary);
    c) Transportation of solids to disposal site.

$$PE_y = PE_{y,ss} + PE_{y,power} + PE_{y,floc,combustion} + PE_{y,transp} \quad (2)$$

Where:

$\mathrm{PE}_{\mathrm{y,ss}}$ = Project emissions from storage, use, destruction or disposal of solids separated in year y (tCO_2e).

$\mathrm{PE}_{\mathrm{y,power}}$ = Project emissions from energy use for operating the separation device in year y (tCO_2e), calculated as per AMS-I.D methods

$\mathrm{PE}_{\mathrm{y,floc,combustion}}$ = Project emissions from combustion of flocculants in year y (tCO_2e)

$\mathrm{PE}_{\mathrm{y,transp}}$ = Project emissions for incremental transportation of matters in the project scenario, beyond the emissions for transportation of solids in the baseline scenario (tCO_2e)

5.2.1 Project emissions from storage, use, destruction or disposal of solids separated ($\mathbf{PE}_{\mathbf{y,ss}}$)

$\mathrm{PE}_{\mathrm{y,ss}}$ is the project emissions from storage, use, destruction or disposal of solids separated in year y (tCO_2e). The project does not store the separated solids on-site and no solids are destructed or disposed either. All the solids from collected swine manure are directly put into fertilizer production through immediate composting following the separation process. Therefore, this section only takes into account the project emissions from the use of solids separated ($\mathrm{PE}_{\mathrm{s,composting,y}}$), which is composting, and so:

$$\mathrm{PE}_{\mathrm{y,ss}} = \mathrm{PE}_{\mathrm{s,composting,y}} \tag{3}$$

According to para.34 of AMS-III.Y, in case the solids separated from manure or from wastewater are composted, the methane emissions from composting of solids separated by the project activity ($\mathrm{PE}_{\mathrm{s,composting,y}}$) is determined as per the relevant procedure in AMS-III.F. According to para.26 of AMS-III.F, $\mathrm{PE}_{\mathrm{s,composting,y}}$ is determined as per Section 6.1 of Tool 13:

$$\mathrm{PE}_{\mathrm{s,composting,y}} = \mathrm{PE}_{\mathrm{EC,y}} + \mathrm{PE}_{\mathrm{FC,y}} + \mathrm{PE}_{\mathrm{CH4,y}} + \mathrm{PE}_{\mathrm{N2O,y}} + \mathrm{PE}_{\mathrm{RO,y}} \tag{4}$$

Where:

$\mathrm{PE}_{\mathrm{s,composting,y}}$ = Project emissions in the year y (tCO_2e)

$\mathrm{PE}_{\mathrm{EC,y}}$ = Project emissions from electricity consumption associated with composting in year y (tCO_2/yr)

$\mathrm{PE}_{\mathrm{FC,y}}$ = Project emissions from fossil fuel consumption associated with composting in year y (tCO_2/yr)

$\mathrm{PE}_{\mathrm{CH4,y}}$ = Project emissions of methane from the composting process in year y (tCO_2e/yr)

$PE_{N2O,y}$ = Project emissions of nitrous oxide from the composting process in year y (tCO_2e/yr)

$PE_{RO,y}$ = Project emissions of methane from run-off wastewater associated with co-composting in year y (tCO_2e/yr)

Determination of the quantity of waste composted ($\mathbf{Q_y}$)

Q_y is the quantity of waste composted in year y (t/yr), which is a parameter required in the determination of emissions from composting. It is determined by monitoring the weight of separated solids delivered to the composting installation using an on-site weighing device as per Option 1, Section 6.1.1.2 of Tool 13.

Determination of composting emissions from electricity consumption ($\mathbf{PE_{EC,y}}$)

$PE_{EC,y}$ is the project emissions from electricity consumption associated with composting in year y (tCO_2/yr). For the project, the composting activity involves electricity consumption from electricity-driven devices for solid fertilizer production. Tool 13 directs the project to calculate $PE_{EC,y}$ using the latest approved version of Tool 05. In particular, the determination of $PE_{EC,y}$ relies on continuously measures rather than non-monitored parameters, aiming for a more accurate assessment of electricity consumption. In accordance with Section 6.2.1 of Tool 05, a monitoring-based approach to calculation composting emissions from consumption of electricity is as follows:

$$PE_{EC,y} = \sum_{j} EC_{PJ,compost,y} \times EF_{EF,compost,y} \times \left(1 + TDL_{compost,y}\right) \tag{5}$$

Where:

$PE_{EC,y}$ = Composting emissions from electricity consumption in year y (tCO_2/yr)

$EC_{PJ,compost,y}$ = Quantity of electricity consumed by composting in year y (MWh/yr)

$EF_{EF,compost,y}$ = Emission factor for electricity generation for composting in year y (tCO_2/MWh). Default value of 1.3 tCO_2/MWh suggested by Tool 05 shall apply

$TDL_{compost,y}$ = Average technical transmission and distribution losses for providing electricity to composting in year y. Default value of 20% suggested by Tool 05 shall apply

For the project, the monitored data for quantity of electricity consumed by composting ($EC_{PJ,compost,y}$) alone is not isolated and thus is unavailable. On the other hand, the electricity consumption from the project activity ($EC_{PJ,project,y}$) that involves all facilities and equipment within the project boundary is collectively measured and recorded. Therefore, to be conservative, $\mathrm{EC_{PJ,project,y}}$ is taken to represent the electricity consumption emissions caused by composting ($\mathrm{EC_{PJ,compost,y}}$), and so:

$$\mathrm{EC_{PJ,compost,y}} = \mathrm{EC_{PJ,project,y}} \tag{6}$$

Determination of composting emissions from fossil fuel consumption ($\mathbf{PE_{FC,y}}$)

$\mathrm{PE_{FC,y}}$ is the project emissions from fossil fuel consumption associated with composting in year y (tCO$_2$/yr). For the project, the composting activity involves fossil fuel consumption from fuel vehicles for transportation and production use. As per Option 2 suggested by Section 6.1.3 of Tool 13, it is calculated as follows:

$$\mathrm{PE_{FC,y}} = \mathrm{Q_y} \times \mathrm{EF_{FC,default}} \tag{7}$$

Where:

$\mathrm{PE_{FC,y}}$ = Project emissions from fossil fuel consumption associated with composting in year y (tCO$_2$/yr)

$\mathrm{EF_{FC,default}}$ = Default emission factor for fossil fuels consumed by the composting activity per tonne of waste (tCO$_2$/t)

Determination of composting emissions of methane ($\mathbf{PE_{CH4,y}}$)

$\mathrm{PE_{CH4,y}}$ is the project emissions of methane from the composting process in year y (tCO$_2$e/yr). As per Section 6.1.4 of Tool 13, it is calculated as follows:

$$\mathrm{PE_{CH4,y}} = \mathrm{Q_y} \times \mathrm{EF_{CH4,y}} \times \mathrm{GWP_{CH4}} \tag{8}$$

Where:

$\mathrm{PE_{CH4,y}}$ = Project emissions of methane from the composting process in year y (tCO$_2$e/yr)

$\mathrm{EF_{CH4,y}}$ = Emission factor of methane per tonne of waste composted valid for year y (t CH$_4$/t).

$\mathrm{GWP_{CH4}}$ = Global Warming Potential of CH$_4$ (tCO$_2$e/tCH$_4$)

Determination of composting emissions of nitrous oxide ($\mathbf{PE_{N20,y}}$)

$\mathrm{PE_{N20,y}}$is the project emissions of nitrous oxide from the composting process in year y (tCO$_2$e/yr). As per Section 6.1.5 of Tool 13, it is calculated as follows:

$$\mathrm{PE_{N20,y}} = \mathrm{Q_y} \times \mathrm{EF_{N2O,y}} \times \mathrm{GWP_{N2O}} \tag{9}$$

Where:

$\mathrm{PE_{N2O,y}}$ = Project emissions of nitrous oxide from composting in year y (tCO_2e/yr)

$\mathrm{EF_{N2O,y}}$ = Emission factor of nitrous oxide per tonne of waste composted valid for year y (tN_2O/t). This is determined following Option 2: Procedure using default values as per Section 6.1.5.2 of Tool 13, where $\mathrm{EF_{N2O,y} = EF_{N2O,default}}$

$\mathrm{GWP_{N2O}}$ = Global Warming Potential of N_2O (tCO_2e/tN_2O)

Determination of composting emissions from run-off wastewater ($\mathbf{PE_{RO,y}}$)

$\mathrm{PE_{RO,y}}$ it the project emissions of methane from run-off wastewater associated with co-composting in year y (tCO_2e/yr). The project does not involve co-composting activities, thus $\mathrm{PE_{RO,y}}$ is zero.

5.2.2 Project emissions from energy use for operating separation devices ($\mathbf{PE_{y,power}}$)

$\mathrm{PE_{y,power}}$ is the project emissions from energy use for operating the separation device in year y (tCO_2e), and is calculated as follows:

$$\mathrm{PE_{y,power} = PE_{y,electricity} + PE_{y,fuel}} \tag{10}$$

According to para.35 of AMS-III.Y, project activity emissions from electricity consumption are determined as per the procedures described in AMS-I.D. The energy consumption of all equipment/devices installed by the project activity, inter alia all equipment to separate solids (including energy used for spray drying and evaporation) shall be included. AMS-I.D calculates CO_2 emissions as follows:

$$\mathrm{PE_{y,electricity} = EG_{PJ,y} \times EF_{grid,y}} \tag{11}$$

For the project, the entire solid separation system is electricity-driven, but the monitored data alone is not isolated. Instead, it is measured collectively with other facilities within the project boundary ($\mathrm{EC_{PJ,project,y}}$). Since Section 4.2.1 has already calculated $\mathrm{EC_{PJ,project,y}}$ using Equation (5) to avoid underestimating composting emissions, $\mathrm{EG_{PJ,y}}$ here is deemed to be zero to avoid double counting, and therefore $\mathrm{PE_{y,electricity}}$ is zero.

This does not underestimate the total electricity consumption within the project boundary, as Equation (5) further considers the average technical transmission and distribution losses for providing electricity to the project (TDL), leading to a more conservative result of project emissions than that of Equation (11).

It is also worth noting that the electricity demand for the project is all supplied by the local power grid that is connected to the Central China Power Grid (CCPG), where the historical emission factors for the recent years are consistently lower than 1.0 tCO_2/MWh[19]. As the default emission factor of 1.3 tCO_2/MWh suggested by Tool 05 is used to calculate $\mathrm{PE_{EC,y}}$ using Equation (5), it in fact assumes a more conservative estimation on project emissions.

The solid separation system is powered by electricity, completely eliminating the need for fossil fuel combustion, thus $\mathrm{PE_{y,fuel}}$ is zero.

5.2.3 Project emissions from combustion of flocculants ($\mathbf{PE_{y,floc,combustion}}$)

$\mathrm{PE_{y,floc,combustion}}$ is the project emissions from combustion of flocculants in year y (tCO_2e). According to para.36 of AMS-III.Y, in case of combustion of separated solids, the CO_2 emissions from the combustion of the non-biomass (i.e. fossil) carbon content of the flocculants shall be considered. In such a case, a default value of 0.2 tCO_2e per tonne of flocculants may be used.

The project does not use flocculants during solid separations, thus $\mathrm{PE_{y,floc,combustion}}$ is zero.

5.2.4 Project emissions from incremental transportation of matters ($\mathbf{PE_{y,trans}}$)

$\mathrm{PE_{y,trans}}$ is the project emissions for incremental transportation of matters in the project scenario, beyond the emissions for transportation of solids in the baseline scenario (tCO_2e). According to para.37 of AMS-III.Y, the incremental transportation of matters for storage, gainful use, destruction and/or land application is calculated as follows:

$$\mathrm{PE_{y,transp}} = \frac{\mathrm{Q_{y,transp}}}{\mathrm{CT_y}} \times \mathrm{DT_y} \times \mathrm{EF_{CO2}} \tag{12}$$

Where:

$\mathrm{Q_{y,transp}}$ = Quantity of matters transported in year y (t)

$\mathrm{CT_y}$ = Average truck capacity for transportation (t/truck)

$\mathrm{DT_y}$ = Average incremental distance for transportation of matters (km/truck)

$\mathrm{EF_{CO2}}$ = CO_2 emission factor for the fuel used for transportation (tCO_2/km, IPCC default values or local values)

For the project, the incremental transportation includes the following:

1) Fresh manure slurry from the swine farms to the project site:

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[19] https://ccer.cets.org.cn/notice/columnDetail?bulletinBoardId=1151185363440635904&bulletinBoardName=%E7%94%B5%E7%BD%91%E5%9F%BA%E5%87%86%E7%BA%BF%E6%8E%92%E6%94%BE%E5%9B%A0%E5%AD%90

All swine farms are located in Leiyang City (county level). Trucks with a minimum capacity of 8 tonnes are commonly used to transport the importing fresh manure slurry.

2) Transportation of separated solids from the on-site solid separation system to the on-site composting system:

According to the FSR, all separated solids undergo immediate on-site composting without the need for interim storage. The transportation distance between on-site solid separation system and the composting system is minimal, typically less than 0.05 km, and can be accomplished using wheelbarrows. Given this short distance and the manual transport method, emissions from transportation are negligible.

3) Transportation of the composted solid fertilizers to local farms for land application:

The resulting compost is used to produce solid fertilizers and subsequently sold to local farmlands for land application across Leiyang City (county level). Trucks with a minimum capacity of 10 tonnes are commonly used to transport the exporting composted solid fertilizers.

5.3 Leakage

The project does not have any leakage emissions to be addressed, due to the following reasons:

1) Solid separation technologies of the project are brand new at the time of commissioning and are not transferred from another activity;

2) The project does not use flocculants during the project activity. However, if flocculants are used and if the flocculants includes ingredients that were manufactured and not a waste product, then leakage emissions must be addressed for that portion of the flocculants. According to para.39 of AMS-III.Y, a default emission factor of 7.9tCO_2e/t may be applied based on the amount of the flocculants that is manufactured (i.e. excluding the portions that obtained from waste sources).

5.4 Estimated Net GHG Emission Reductions and Removals

Emission reductions are calculated as follows:

$$\mathrm{ER_y} = \mathrm{BE_y} - \mathrm{PE_y} - \mathrm{LE_y} \tag{13}$$

Where:

$\mathrm{ER_y}$ = Emission reductions during the year y (tCO_2)

$\mathrm{BE_y}$ = Baseline emissions during the year y (tCO_2)

PE_y = Project emissions during the year y (tCO_2)

LE_y = Leakage during the year y (tCO_2)

Based on the calculation methods stated above, the ex-ante GHG emission reductions are calculated as follows:

Ex-ante Calculation of Baseline Emissions

The ex-ante calculation of the baseline emissions (BE_y) are calculated as per Equation (1), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$B_{o,w,y}$	0.45	m^3CH_4/kg-VS	Default, IPCC 2019 refinement, Table 10.16A, Swine, Other Regions, High productivity systems. The project only involves manure from swine ($B_{o,w,y} = B_{o,swine,y}$).
$M_{ss,y}$	21,000,000	kg	Feasibility Study Report
$VS_{ss,y}$	0.55	kg/kg	Feasibility Study Report
UF_b	0.94	-	Default, AMS-III.Y
GWP_{CH4}	28.0	tCO_2e/tCH_4	Default, IPCC AR5
D_{CH4}	0.67	kg/m^3	Default, AMS-III.Y
$MS_{Bl,i}$	100%	-	Default, AMS-III.Y. There was only one type of treatment system existed prior to the project implementation, as per the Feasibility Study Report.
$MCF_{b,i}$	73%	-	Default, IPCC 2019 refinement, Table 10.17, Uncovered anaerobic lagoon. The project falls within warm, temperate, moist climate zone. There was only one type of treatment system existed prior to the project implementation, as per the Feasibility Study Report.
BE_y	66,907	tCO_2e	Calculated

Ex-ante Calculation of Project Emissions

The ex-ante calculation of the project emissions (PE_y) are calculated as per Equation (2), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$PE_{y,ss}$	6,935	tCO_2e	Calculated
$PE_{y,power}$	-	tCO_2e	-
$PE_{y,floc,combustion}$	-	tCO_2e	-
$PE_{y,transp}$	2,665	tCO_2e	Calculated
PE_y	9,600	tCO_2e	Calculated

1) The ex-ante calculation of the project emissions from storage, use, destruction or disposal of solids separated ($PE_{y,ss}$) are calculated as per Equation (3) and Equation (4), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$PE_{EC,y}$	125	tCO_2e	Calculated
$PE_{FC,y}$	1,087	tCO_2e	Calculated
$PE_{CH4,y}$	2,940	tCO_2e	Calculated
$PE_{N2O,y}$	2,783	tCO_2e	Calculated
$PE_{RO,y}$	-	tCO_2e	Irrelevant
$PE_{y,ss}$ or $PE_{s,composting,y}$	6,935	tCO_2e	Calculated

a) The ex-ante calculation of the composting emissions from electricity consumption ($PE_{EC,y}$) are calculated as per Equation (5) and Equation (6), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$EC_{PJ,compost,y}$ or $EC_{PJ,project,y}$	80	MWh/yr	Feasibility Study Report
$EF_{EF,compost,y}$	1.3	tCO_2e/MWh	Default, Tool 5
$TDL_{compost,y}$	20%	-	Default, Tool 5
$PE_{EC,y}$	125	tCO_2e/yr	Calculated

b) The ex-ante calculation of the composting emissions from fossil fuel consumption ($\mathrm{PE_{FC,y}}$) are calculated as per Equation (7), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$\mathrm{EF_{FC,default}}$	0.0207	tCO_2/t	Default, Tool 13
$\mathrm{Q_y}$	52,500	t/yr	Feasibility Study Report
$\mathrm{PE_{FC,y}}$	1,087	tCO_2e/yr	Calculated

c) The ex-ante calculation of the composting emissions of methane ($\mathrm{PE_{CH4,y}}$) are calculated as per Equation (8), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$\mathrm{EF_{CH4,y}}$	0.002	tCH_4/t	Default, Tool 13
$\mathrm{GWP_{CH4}}$	28	tCO_2e/tCH_4	Default, IPCC AR5
$\mathrm{Q_y}$	52,500	t/yr	Feasibility Study Report
$\mathrm{PE_{CH4,y}}$	2,940	tCO_2e/yr	Calculated

d) The ex-ante calculation of the composting emissions of nitrous oxide ($\mathrm{PE_{N20,y}}$) are calculated as per Equation (9), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$\mathrm{EF_{N2O,y}}$	0.0002	tN_2O/t	Default, Tool 13
$\mathrm{GWP_{N2O}}$	265	tCO_2e/tN_2O	Default, IPCC AR5
$\mathrm{Q_y}$	52,500	t/yr	Feasibility Study Report
$\mathrm{PE_{N20,y}}$	2,783	tCO_2e/yr	Calculated

e) The ex-ante calculation of the projection emissions from incremental transportation of matters ($\mathrm{PE_{y,transp}}$) are calculated as per Equation (12), incorporating the following specified parameters:

Parameter	Ex-ante Value	Unit	Sources
$\mathrm{Q_{y,transp}}$	130,000	t	Feasibility Study Report
$\mathrm{CT_y}$	8	t/truck	Procurement contract

DT_y	200	km/truck	Maps
EF_{CO2}	0.00082	tCO_2e/km	Default, Local values & IPCC
$PE_{y,transp}$	2,665	tCO_2e	Calculated

Ex-ante Calculation of Emission Reductions

The ex-ante calculation of the emission reductions (ER_y) are calculated as per Equation (13). The results are shown in the table below. Since the crediting period begins and ends part-way through 2022 and 2029, these years' total reductions are adjusted (10 months of credit in 2022, 3 months of credit in 2029). It is estimated that for the 1st crediting period (28-Mar-2022 to 27-Mar-2029), the total GHG emission reductions will reach to 401,149tCO_2e, with annual emission reductions of 57,307tCO_2e.

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
28-Mar-2022 to 31-Dec-2022	51,143	7,338	0	43,805
01-Jan-2023 to 31-Dec-2023	66,907	9,600	0	57,307
01-Jan-2024 to 31-Dec-2024	66,907	9,600	0	57,307
01-Jan-2025 to 31-Dec-2025	66,907	9,600	0	57,307
01-Jan-2026 to 31-Dec-2026	66,907	9,600	0	57,307
01-Jan-2027 to 31-Dec-2027	66,907	9,600	0	57,307
01-Jan-2028 to 31-Dec-2028	66,907	9,600	0	57,307
01-Jan-2029 to 27-Mar-2029	15,764	2,262	0	13,502
Total	468,349	67,200	0	401,149

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	$B_{o,w,y}$
Data unit	m^3CH_4 per kg of VS

Description	Weighted methane-producing potential of the volatile solids separated by the project in year y
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	0.45 for swine in high productivity systems in other regions
Justification of choice of data or description of measurement methods and procedures applied	The project only collects manure from swine, thus $\mathrm{B_{o,w,y}} = \mathrm{B_{o,swine,y}}$. No country or regional specific value is available, thus the default value of Other Regions from Tables 10.16A of Chapter 10, Volume 4, 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories is applied The swine farms are 100 percent market oriented with high level of capital input and high level of overall herd (flock) performance. Feed is purchased from local markets, swine are improved through breeding practices for commercial production and thereby the swine farms are with “High Productivity Systems”, and the values of high productivity systems are applied
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\mathrm{UF_b}$
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.Y. (Reference: FCCC/SBSTA/2003/10/Add.2, page 25)
Value applied	0.94
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\mathrm{GWP_{CH4}}$
Data unit	tCO_2e/tCH_4

Description	Global Warming Potential of methane applicable to the crediting period
Source of data	IPCC Fifth Assessment Report
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of baseline and project emissions
Comments	Shall be updated according to any future revision to VCS standard by VERRA

Data / Parameter	GWP_{N2O}
Data unit	tCO_2e/tN_2O
Description	Global Warming Potential of nitrous oxide applicable to the crediting period
Source of data	IPCC Fifth Assessment Report
Value applied	265
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of project emissions
Comments	Shall be updated according to any future revision to VCS standard by VERRA

Data / Parameter	D_{CH4}
Data unit	kg per m^3 at 20 °C and 1 atm pressure
Description	Conversion factor of m^3 CH_4 to kilograms
Source of data	AMS-III.Y
Value applied	0.67

Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$MS_{Bl,i}$
Data unit	-
Description	Fraction of manure handled in the baseline anaerobic manure management system i (fraction, mass basis)
Source of data	AMS-III.Y
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	Values should be based on historical information from previous year(s) before project. If historical information is not available, the capacity (t/day) of each baseline management system shall be verified before the project start. During the project, it would be assumed that the management systems with lower MCFs would be used up to their full capacity, and only thereafter the systems with larger MCF would be used. The value of $MS_{Bl,i}$ is “1.0” if only one type of treatment system was used in the baseline The agency responsible for preparing the FSR conducted a thorough baseline survey during the preparation stage to understand the existing conditions and practices at that time. Based on historical records documented in the FSR prepared in Jan-2020, nearly 2 years before the project implementation date of 28-Mar-2022, there was only one type of treatment system existed prior to the project implementation, where animal manure was left to decay in uncovered anaerobic lagoon at the local swine farms. Therefore, $MS_{Bl,i}$ is 1.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$MCF_{b,i}$
Data unit	-

Description	Methane conversion factor for the baseline anaerobic manure management system
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	73%
Justification of choice of data or description of measurement methods and procedures applied	The agency responsible for preparing the FSR conducted a thorough baseline survey during the preparation stage to understand the existing conditions and practices at that time. Based on historical records documented in the FSR prepared in Jan-2020, nearly 2 years before the project implementation date of 28-Mar-2022, there was only one type of treatment system existed prior to the project implementation, where animal manure was left to decay in uncovered anaerobic lagoon at the local swine farms As no country or regional specific value is available, default value from Table 10.17 of Chapter 10, Volume 4, 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories is applied. The project falls within warm, temperate, moist climate zone based off Figure 10A.1 of the same chapter. Consequently, the corresponding annual MCF of uncovered anaerobic lagoons in this climate zone is 73%.
Purpose of Data	Calculation of baseline emissions
Comments	Shall be updated according to any future revision to VCS standard by VERRA

Data / Parameter	$\mathrm{EF_{EF,compost,y}}$
Data unit	tCO_2/MWh
Description	Emission factor for electricity generation for composting in year y
Source of data	Tool 05
Value applied	1.3
Justification of choice of data or description of measurement methods and procedures applied	As per Tool 05, the determination of the emission factors for electricity generation ($\mathrm{EF_{EF,j/k/l}}$) in the project scenario depends on which scenario (A, B or C), as described in Section 2.2, paragraph 5 of Tool 05 that applies to the source of electricity consumption that would be displaced in the baseline by electricity generated in the project. This project is considered as Scenario A

	as it consumes electricity from the power grid, and Tool 05 suggests two options of calculation: 1) Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the “Tool to calculate the emission factor for an electricity system” ($EF_{EF,j/k/l}$= $EF_{grid,CM,y}$); or 2) Option A2: Use the following conservative default values: a) A value of 1.3 tCO$_2$/MWh if: i. Scenario A applies only to project and/or leakage electricity consumption sources but not to baseline electricity consumption sources; or ii. Scenario A applies to: both baseline and project (and/or leakage) electricity consumption sources; and the electricity consumption of the project and leakage sources is greater than the electricity consumption of the baseline sources; b) A value of 0.4 tCO$_2$/MWh for electricity grids where hydro power plants constitute less than 50% of total grid generation in average of the five most recent years, or based on long-term averages for hydroelectricity production, and a value of 0.25 t CO$_2$/MWh for other electricity grids. These values can be used if: i. Scenario A applies only to baseline electricity consumption sources but not to project or leakage electricity consumption sources; or Scenario A applies to: both baseline and project (and/or leakage) electricity consumption sources; and the electricity consumption of the baseline sources is greater than the electricity consumption of the project and leakage sources. The project applies Option A2 to determine $EF_{EF,comp,y}$ using the conservative default values. As the project scenario applies only to project electricity consumption sources but not to baseline electricity consumption sources, $EF_{EF,comp,y}$ = 1.3 tCO$_2$/MWh.
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$TDL_{compost,y}$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to composting in year y

Source of data	Tool 05
Value applied	20%
Justification of choice of data or description of measurement methods and procedures applied	As per Tool 05, in case of other scenarios (scenario A and scenario C, cases C.I and C.III), choose one of the following options: 1) Use annual average value based on the most recent data available within the host country; 2) Use as default values of 20% for: a) project or leakage electricity consumption sources; b) baseline electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is larger than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies; 3) Use as default values of 3% for: a) baseline electricity consumption sources; b) project and leakage electricity consumption sources if the electricity consumption by all project and leakage electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies is smaller than the electricity consumption of all baseline electricity consumption sources to which scenario A or scenario C (cases C.I or C.III) applies No country or regional specific value is available. As the project scenario applies only to project electricity consumption sources, the default value of 20% is applied
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$EF_{FC,default}$
Data unit	tCO_2/t
Description	Default emission factor for fossil fuel consumed by the composting activity per tonne of waste composted (wet basis)
Source of data	Tool 13. The emission factor was selected based on a review of fossil fuel consumption per tonne of waste composed in relevant validation reports of CDM projects using a conservative default emission factor for diesel (from the 2006 IPCC Guidelines)

Value applied	0.0207
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$EF_{CH4,y}$
Data unit	tCH_4/t
Description	Emission factor of methane per tonne of waste composted valid for year y
Source of data	Tool 13. The emission factor was selected based on studying published results of emission measurements from composting facilities, literature reviews on the subject and published emission factors. Data from recent, high quality sources was analyzed and a value conservatively selected from the higher end of the range in results
Value applied	0.002
Justification of choice of data or description of measurement methods and procedures applied	As per Tool 13, there are two options which the PP may choose for determining $EF_{CH4,y}$: 1) Option 1: Procedure using monitored data $EF_{CH4,y}$ is determined based on measurements of the methane emissions during a composting cycle 2) Option 2: Procedure using default values A default value is used: $EF_{CH4,y} = EF_{CH4,default}$, which is the default emission factor of methane per tonne of waste composted (wet basis), as per "Data and parameters not monitored" section of Tool 13 The project chooses Option 2 for determining $EF_{CH4,y}$
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$EF_{N2O,y}$

Data unit	tN_2O/t
Description	Emission factor of nitrous oxide per tonne of waste composted valid for year
Source of data	As per “Data and parameters not monitored” section of Tool 13, the emission factor was selected based on studying published results of emission measurements from composting facilities, literature reviews on the subject and published emission factors. Data from recent, high quality sources was analyzed and a value conservatively selected from the higher end of the range in results
Value applied	0.0002
Justification of choice of data or description of measurement methods and procedures applied	As per Tool 13, there are two options which the PP may choose for determining $EF_{N2O,y}$: 1) Option 1: Procedure using monitored data $EF_{N2O,y}$ is determined based on measurements of the methane emissions during a composting cycle 2) Option 2: Procedure using default values A default value is used: $EF_{N2O,y} = EF_{N2O,default}$, which is the default emission factor of nitrous oxide per tonne of waste composted (wet basis), as per “Data and parameters not monitored” section of Tool 13 This project chooses Option 2 for determining $EF_{N2O,y}$
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$EF_{N2O,y}$
Data unit	tN_2O/t
Description	Emission factor of nitrous oxide per tonne of waste composted valid for year
Source of data	As per “Data and parameters not monitored” section of Tool 13, the emission factor was selected based on studying published results of emission measurements from composting facilities, literature reviews on the subject and published emission factors. Data from recent, high quality sources was analyzed and a value conservatively selected from the higher end of the range in results
Value applied	0.0002

Justification of choice of data or description of measurement methods and procedures applied	As per Tool 13, there are two options which the PP may choose for determining $\mathrm{EF_{N2O,y}}$: 3) Option 1: Procedure using monitored data $\mathrm{EF_{N2O,y}}$ is determined based on measurements of the methane emissions during a composting cycle 4) Option 2: Procedure using default values A default value is used: $\mathrm{EF_{N2O,y}} = \mathrm{EF_{N2O,default}}$, which is the default emission factor of nitrous oxide per tonne of waste composted (wet basis), as per “Data and parameters not monitored” section of Tool 13 This project chooses Option 2 for determining $\mathrm{EF_{N2O,y}}$
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$\mathrm{DT_y}$
Data unit	km/truck
Description	Average incremental distance for transportation of matters
Source of data	Maximum road distance of incremental transportation obtained from the maps
Value applied	200
Justification of choice of data or description of measurement methods and procedures applied	The incremental transportation of the project includes the import of fresh manure slurry from the swine farms to the project site, as well as the export of the solid fertilizer from the project site to the local farmlands. Both swine farms and local farmlands are located within Leiyang City (county level). To ensure conservative calculations, the maximum road distance from the project site to the farthest point within the county boundary has been considered for both import and export activities. The farthest location from the project site is identified at a latitude of 26.693° N and a longitude of 113.177° E using ArcMap. The road distance to this location is confirmed to be 85.8km through Google Maps and is rounded to 100km to provide a more conservative estimate. This results in a total incremental transportation distance of 200km for the project, accounting for both the import of manure slurry and the export of solid fertilizer.
Purpose of Data	Calculation of project emissions

Comments	-

Data / Parameter	CT_y
Data unit	tonnes/truck
Description	Average truck capacity for transportation
Source of data	Minimum capacity of transportation trucks
Value applied	8
Justification of choice of data or description of measurement methods and procedures applied	The incremental transportation of the project includes the import of fresh manure slurry from the swine farms to the project site, as well as the export of the solid fertilizer from the project site to the local farmlands. The capacities of transportation trucks vary, with a minimum of 8 tonnes/truck for imports and 10 tonnes/truck for exports, as specified in procurement contrasts and usage instructions. To ensure conservative calculations, the minimum capacity of 8 tonnes/truck has been considered for both import and export activities.
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	EF_{CO2}
Data unit	tCO_2/km
Description	CO_2 emission factor for the fuel used for transportation (IPCC default values or local values)
Source of data	Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial), 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	0.00082
Justification of choice of data or description of measurement methods and procedures applied	The incremental transportation for the project is carried out by diesel trucks. The CO_2 emission factor for fuel vehicle based off distances travelled is calculated as follows: $EF_{CO2} = (V \times \rho) \times NCV \times EF_{CO2,TJ}$ The fuel consumption per 100 km (V) for diesel trucks is derived from the "Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China

	(Trial)", organized by the National Development and Reform Commission. For conservative calculations, the project uses the smallest load truck (8 tonnes, as detailed in CT_y), with a corresponding fuel consumption of 0.0307 m^3/100km according to the "Guidelines" The default density (ρ) for diesel oil is 0.84 t/m^3, also obtained from the "Guidelines" The default net calorific value (NCV) of diesel oil is 43 TJ/Gg obtained from Table 1.2 of the Introduction Chapter, Volume 2, the 2006 IPCC Guidelines for National Greenhouse Gas Inventories The road transportation default CO_2 emission factor of diesel oil ($EF_{CO2,TJ}$) is 74,100 kgCO_2/TJ obtained from Table 3.2.1 of Chapter 3, Volume 2, the 2006 IPCC Guidelines for National Greenhouse Gas Inventories As such: $EF_{CO2} = (0.0307\ m^3/100km \times 0.84\ t/m^3) \times 43\ TJ/Gg \times 74{,}100\ kgCO_2/TJ = 0.00082\ tCO_2/km$
Purpose of Data	Calculation of project emissions
Comments	-

6.2 Data and Parameters Monitored

Data / Parameter	$M_{ss,y}$
Data unit	kg
Description	Mass (dry matter basis) of total separated solids in year y
Source of data	Ex-ante values are historical records documented in the FSR prepared in Jan-2020, nearly 2 years before the project implementation date of 28-Mar-2022 Ex-post values are from the measuring records
Description of measurement methods and procedures to be applied	$M_{ss,y}$ on a dry matter basis is determined based on both the wet weight and the dry matter content of the separated solids: a) The wet weight of the separated solids is determined based on the amount of waste subjected to the composting process (Q_y), which is measured continuously by direct weighing using the scale installed near the separation system. The weighing records are aggregated by day In the context of the project, all separated solids are composted on site prior to any secondary use. As composting becomes the

	exclusive way of utilizing the separated solids, the wet weight of the separated solids essentially equates to Q_y, and therefore: $\text{Wet weight of separated solids (kg)} = Q_y(t) \times 1000\ \text{kg}$ b) The dry matter content of the separated solids is determined through laboratory testing The staff first collects representative samples from the separated solids, bringing them to the laboratory for further analysis. In the laboratory, samples are placed in a laboratory oven where free water is evaporated until a consistent weight is achieved. An analytical balance is employed for the precise measurement of samples and containers. The difference between sample weights before and after the drying process reflects the moisture content of the separated solid. Then, the dry matter content of the separated solids can be calculated as: $\text{Dry matter content (\%)} = 1 - \text{Moisture content (\%)}$ As samples are collected and analyzed on a daily basis, there should be no requirement for additional sampling procedures The staff records Q_y, the moisture content and the dry matter content to the measuring records for subsequent determination of the monthly $M_{ss,y}$. Details on calculation methods are provided in the "Calculation Method" section below
Frequency of monitoring/recording	Monthly
Value applied	Ex-ante: 21,000,000 (Wet weight or Q_y: 52,500 t; Moisture content: 60%)
Monitoring equipment	Scale, laboratory oven, analytical balance
QA/QC procedures to be applied	Monitoring equipment should be calibrated by an accredited organization according to national standards, specifically the Verification Regulation for Digital Indicating Weighing Instruments (JG 539-2016) and the Verification Regulation of Electronic Balance (JG1036-2008), by qualified tried parties
Purpose of data	Calculation of baseline emissions
Calculation method	$M_{ss,y}\ \text{(kg)} = \text{Wet weight of separated solids (kg)} \times \text{Dry matter content (\%)}$ $= Q_y\ (t) \times 1000\ \text{kg} \times \left(1 - \text{Moisture content (\%)}\right)$
Comments	If the dry matter content of a sample is lower than the minimum value of 20%, no emission avoidance will be assigned to the amount of separated solids from which the sample is representative

Data / Parameter	$VS_{ss,y}$
Data unit	kg/kg
Description	Volatile solids content of the separated solids in year y on a dry matter basis
Source of data	Ex-ante values are historical records documented in the FSR prepared in Jan-2020, nearly 2 years before the project implementation date of 28-Mar-2022 Ex-post values are from the measuring records
Description of measurement methods and procedures to be applied	$VS_{ss,y}$ is determined through laboratory testing following the procedures in Annex 2 of AM0073 (Version 01)[20]. In the context of the project, representative samples from the separated solids have been collected to determine the dry matter content of $M_{ss,y}$ in the laboratory. The solid residues are then placed in Muffle furnace at 600ºC for at least 1 hour. The difference between sample weights before and after the furnacing process reflects the volatile solid content of the separated solids, as depicted in the formula below: $$\text{Volatile matter (dry basis)} = \frac{W_2 - W_f}{W_2 - W_1}$$ Where W_1 is the weight of sample container, W_2 is combined weight of the sample container and oven dried sample, W_f is the combined constant weight of the sample container and sample after heating at 600ºC The staff records the calculated $VS_{ss,y}$ to the measuring records As samples are collected and analyzed on a daily basis, there should be no requirement for additional sampling procedures
Frequency of monitoring/recording	Yearly
Value applied	Ex-ante: 0.55
Monitoring equipment	Muffle furnace, analytical balance
QA/QC procedures to be applied	Monitoring equipment should be calibrated by an accredited organization according to national standards, specifically the Verification Regulation for Digital Indicating Weighing Instruments (JG 539-2016) and the Verification Regulation of Electronic Balance (JG1036-2008), by qualified tried parties

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[20] https://cdm.unfccc.int/UserManagement/FileStorage/CDMWF_AM_8CTM5MFZLTDO7SZK1A6D7AK3YPIG7S

Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	Q_y
Data unit	t/yr
Description	Quantity of waste composted in year y (wet basis)
Source of data	Ex-ante values are obtained from the Feasibility Study Report Ex-post values are from the measuring records
Description of measurement methods and procedures to be applied	As per Tool 13, there are two options to determine Q_y: 1) Option 1: Procedure using a weighing device Monitor the weight of waste delivered to the composting installation using an on-site weighbridge or any other applicable and calibrated weighing device (e.g. belt-scales) 2) Option 2: Procedure without using a weighing device This procedure shall only be applied in the case that there is no weighbridge or any other applicable and calibrated weighing device available on site. Under this procedure, Q_y is calculated based on the carrying capacity of each truck delivering waste to the composting installation in year y (CT_y) The project chooses Option 1 for determining Q_y. The weight of waste delivered to the composting system can be measured on daily basis by direct weighing using the scale installed near the separation system, and the data is then aggregated into the measurement records
Frequency of monitoring/recording	Daily
Value applied	Ex-ante: 52,500
Monitoring equipment	Same scale as that used for measuring $M_{ss,y}$
QA/QC procedures to be applied	Following the same procedure established for $M_{ss,y}$
Purpose of data	Calculation of project emissions
Calculation method	-

Comments	-

Data / Parameter	$\mathrm{EC}_{\mathrm{PJ,compost,y}}$
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption from composting in year y
Source of data	Ex-ante values are obtained from the Feasibility Study Report Ex-post values are from the monthly reading records
Description of measurement methods and procedures to be applied	Continuous measurements from on-site electricity meters are utilized to calculate the electricity consumption from composting, as it provides a more accurate output than non-monitored parameters In the context of the project, a main electricity meter is installed on-site to collectively measure the electricity consumption from the project activity ($\mathrm{EC}_{\mathrm{PJ,project,y}}$) that involves all facilities and equipment within the project boundaries. The monitored data for composting ($\mathrm{EC}_{\mathrm{PJ,compost,y}}$) alone is not isolated and thus is unavailable. To avoid underestimating the project emissions, $\mathrm{EC}_{\mathrm{PJ,project,y}}$ is employed to represent $\mathrm{EC}_{\mathrm{PJ,compost,y}}$, and so: $\mathrm{EC}_{\mathrm{PJ,compost,y}} = \mathrm{EC}_{\mathrm{PJ,project,y}}$ At the end of each billing period, the staff read the electricity meter and record the quantity of electricity consumption to the monthly reading records
Frequency of monitoring/recording	Monthly
Value applied	Ex-ante: 80
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Monitoring equipment installed on-site should be calibrated by an accredited organization according to national standards The metering records is crosschecked against the payment receipts
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$N_{LT,y}$
Data unit	Number
Description	Number of swine for the year y
Source of data	Ex-ante values are historical records of the number of swine on hand, as documented in the FSR prepared in Jan-2020, nearly 2 years before the project implementation date of 28-Mar-2022 Ex-post values are from the annual operation records
Description of measurement methods and procedures to be applied	Following procedures in AMS-III.D, $N_{LT,y}$ is calculated as follow: $$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$$ Where $N_{da,y}$ is the number of days animal is alive in the farm in the year y, while $N_{p,y}$ is the number of animals produced annually of type LT for the year y. Both are recorded and aggregated to annual operation records by responsible staff on each swine farm. The PP collects the annual operation records and calculate $N_{LT,y}$
Frequency of monitoring/recording	Yearly
Value applied	Ex-ante: 84,745 (number of swine on hand in 2019)
Monitoring equipment	-
QA/QC procedures to be applied	As per AMS-III.D, the consistency between the value and indirect information should be assessed
Purpose of data	Fulfilling monitoring requirements of para. 43(b) of AMS-III.Y to achieve a more comprehensive monitoring plan. This parameter does not affect the ER calculations
Calculation method	-
Comments	-

Data / Parameter	$VS_{LT,y}$
Data unit	kg per year of VS excreted by swine
Description	Annual amount of volatile solids excreted by swine managed by the management system in year y

Source of data	Ex-ante values are determined following procedures in AMS-III.D, which suggests that if country specific VS values are not available, IPCC default values can be used Ex-post values are updated as per VCS Standard's requirements
Description of measurement methods and procedures to be applied	The project only involves swine farms, thus $\mathrm{VS}_{\mathrm{LT,y}} = \mathrm{VS}_{\mathrm{swine,y}}$ The swine farms are 100 percent market oriented with high level of capital input and high level of overall herd (flock) performance. Feed is purchased from local markets, swine are improved through breeding practices for commercial production and thereby the swine farms are with "High Productivity Systems", and the values of high productivity systems are applied Table 10.13A of Chapter 10, Volume 4, 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories provides default daily $\mathrm{VS}_{\mathrm{swine}}$ for various swine categories in high productivity systems in Asia, including feeding and breeding swine. As the swine farms contain various swine types, the default value of 4.3/1000kg animal mass/day is applied Meanwhile, Table 10A.5 provides default live weight values for various categories of swine. The default value of 69kg animal mass/hd is applied Therefore, $\mathrm{VS}_{\mathrm{swine,y}}$ is calculated as: $\mathrm{VS}_{\mathrm{swine,y}} = 4.3\mathrm{kg}/1000\mathrm{kg\ animal\ mass/day} \times 69\mathrm{kg\ animal\ mass}/\mathrm{hd} \times 365\mathrm{days} = 108.30\mathrm{kg/hd/y}$ Values should be updated according to any future revision to VCS Standard by VERRA
Frequency of monitoring/recording	Yearly
Value applied	Ex-ante: 108.30 for swine in high productivity systems in Asia
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Fulfilling monitoring requirements of para. 43(b) of AMS-III.Y to achieve a more comprehensive monitoring plan. This parameter does not affect the ER calculations
Calculation method	-
Comments	-

Data / Parameter	$Q_{y,transp}$
Data unit	tonnes
Description	Quantity of matter transported in year y
Source of data	Ex-ante values are documented in the FSR prepared in Jan-2020, nearly 2 years before the project implementation date of 28-Mar-2022 Ex-post values are from the inventory records
Description of measurement methods and procedures to be applied	The incremental transportation of the project includes the internal ($Q_{y,int}$) and external ($Q_{y,ext}$) transportation: $Q_{y,transp} = Q_{y,internal} + Q_{y,external}$ $Q_{y,internal}$ is the transportation of separated solids from the on-site solid separation system to the on-site composting system. all separated solids undergo immediate on-site composting without the need for interim storage. The transportation distance between on-site solid separation system and the composting system is minimal, typically less than 0.05 km, and can be accomplished using wheelbarrows. Given this short distance and the manual transport method, emissions from transportation are negligible $Q_{y,external}$ includes the transportation of the import of fresh manure slurry from the swine farms to the project site, as well as the export of the solid fertilizer from the project site to the local farmlands. Both quantities could be measured on daily basis by direct weighing using the scale installed near the separation system, and the data is then aggregated into the inventory records
Frequency of monitoring/recording	Yearly
Value applied	Ex-ante: 130,000 ($Q_{y,internal}$ is 0, $Q_{y,external}$ includes 100,000 of fresh manure slurry and 30,000 of solid fertilizers)
Monitoring equipment	Same scale as that used for measuring $M_{ss,y}$
QA/QC procedures to be applied	Following the same procedure established for $M_{ss,y}$
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

6.3 Monitoring Plan

The monitoring plan presented in this JPM assures that real, measurable, long-term GHG emission reductions can be monitored, recorded, and reported. It is a crucial procedure to identify the final VCUs of the project. This monitoring plan has been implemented by the project owner during the 1st monitoring period. The details of the monitoring plan are specified as follows:

1) Data and parameters monitoring methodology

Data and parameters available at validation can be retrieved from Section 6.1. Data and parameters to be monitored throughout the CP are aligned with Section 6.2 of this repot and are listed below. Also refer to Figure 3.1 for the location of monitoring points.

Monitoring Point	Parameter	Description of the project
E1	$M_{ss,y}$	The wet weight of the separated solids equates to the amount of waste subjected to the composting process (Q_y). The dry matter content moisture content is determined based on the moisture content which is measured by the laboratory oven and the analytical balance in the laboratory. The staff should record Q_y, the moisture content and the dry matter content to the measuring records for subsequent determination of the monthly $M_{ss,y}$.
E2	$VS_{ss,y}$	The volatile solids content of the separated solids is determined through laboratory testing following the procedures in Annex 2 of AM0073 (Version 01). The staff should record the $VS_{ss,y}$ to the measuring records.
E1	Q_y	The quantity of waste composted is measured on daily basis by direct weighing using the scale installed near the separation system. The staff should record Q_y to the measuring records.
E3	$EC_{PJ,compost,y}$	The electricity consumption within the project boundaries ($EC_{PJ,project,y}$), as measured by on-site electricity meters, represents the electricity consumption associated with the composting process. The staff should read the electricity meters and record the quantity of electricity consumption to the monthly reading records. The reading records should be crosschecked against the payment receipts.

Swine farm	$N_{LT,y}$	The number of swine on farms. The number of swine produced annually and the days they are alive in the farm are both recorded and aggregated to annual operation records by responsible staff on each swine farm. The PP collects the annual operation records and calculate $N_{LT,y}$. The consistency between the value and indirect information should be assessed
Swine farm	$VS_{LT,y}$	Annual amount of volatile solids excreted by swine managed by the management system. Values should be updated according to any future revision to VCS Standard by VERRA.
Project site	$Q_{y,transp}$	The quantity of matters transported in year y. It should be measured on daily basis by direct weighing using the scale installed near the separation system, and the data is then aggregated into the inventory records.

Operational conditions of the manure management system confirmed at validation are listed below:

Condition	Description of the project
Animals confinement management status	Swine have been managed in confined conditions, entailing the use of enclosed building structures equipped with ventilation systems, lighting facilities, and temperature control devices to ensure good air quality, appropriate temperatures, and adequate lighting conditions. Cement floors or manure leakage boards are commonly featured to provide a neat environment. Swine farms are equipped with fire prevention equipment, emergency exits, and surveillance cameras. Information was obtained from FSR.
Anaerobic lagoon overflow discharge status	Swine manure from anaerobic lagoons were ultimately utilized for land application in Leiyang City (county level), rather than being discharged into river/lake/sea. Information was obtained from FSR and historical operation records of baseline lagoons.
Solids retention time and removal operation interval	Since the decomposed manure eventually applied to farmlands in Leiyang City (county level), farmers adhered to a schedule of removing solids from the anaerobic lagoons every 6 months in minimum to ensure thorough decomposition, safeguarding against adverse effects on the crops. Information was obtained from FSR and historical operation records of baseline lagoons.

Solids separation facilities quality control	A quality control plan has been developed by the PP. It includes regular inspections and maintenance, dedicated on-site monitoring by trained personnel, and prompt repairs as needed. Regular inspections will ensure all equipment components are functioning optimally, while dedicated on-site monitoring by trained personnel will provide real-time oversight of operational parameters. Any deviations or issues detected will be addressed promptly through timely repairs to ensure uninterrupted and efficient operation of the solids separation facilities. Information was obtained from FSR and equipment maintenance records of solid-liquid separators dating back to the start of the project.

2) Management Structure

The project owner should organize a specific VCS team in the project development department to be responsible for data collection, supervision, and witness to the whole process of data monitoring and recording. A VCS manager should be appointed to take full responsibility for the overall monitoring of the project. The monitoring and recording of parameters listed in Section 6.2 of this JPM should be carried out by designated monitoring officers. In addition, the project developer should appoint internal verifiers who is responsible for internal check of the measurement, collection of relevant record, and the calculation of the emission reductions. A monitoring and management manual of the project that identifies detailed duties and responsibilities of the relevant parties should be developed and served as the basis of the project monitoring. Figure 5.1 shows the operation and management structure of the project:

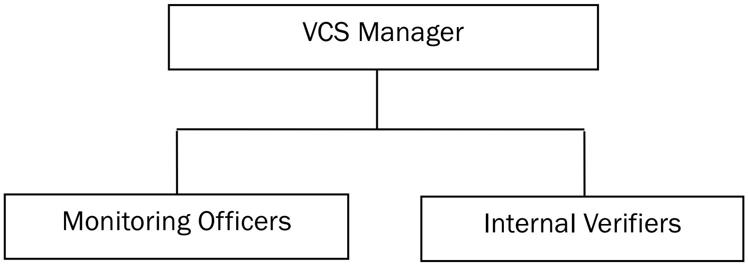

Figure 5-1 Operation and management structure of the project

3) Data collection

Monitoring officers should be responsible for data collection. Designated teams should monitor, collect and record the monitored data regularly. The relevant production and operation data records should serve as the main data source for emission reduction calculation. All record files should be collected by a designated monitoring officer, who should prepare backup in time and archive all documents properly.

4) QA&QC

Measuring instruments should be calibrated by an accredited organization according to national standards.

5) Data file management

All monitoring data should be filed regularly, and the data files should be archived in printed hard copy or electronic disk copy. Other documents in paper, such as maps, forms, and environment assessment reports, should be preserved as well. All monitoring data collected should be archived electronically, and the original copies should be retained for a minimum of 2 years after the end of the crediting period. The project owner should provide original records and documents if necessary.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$M_{ss,y}$
Data unit	kg
Description	Mass (dry matter basis) of total separated solids in year y
Value applied:	Retrieved from the measuring records of the 1st monitoring period and aggregated on a calendar year basis: <table><tr><th>Vintage</th><th>$M_{ss,y}$</th></tr><tr><td>2022 (28-Mar-2022 to 31-Dec-2022)</td><td>16,093,939</td></tr><tr><td>2023 (01-Jan-2023 to 30-Sep-2023)</td><td>15,184,739</td></tr><tr><td>Total</td><td>31,278,678</td></tr></table> Values above are calculated as follows: $M_{ss,y}\ (\text{kg}) = \text{Wet weight of separated solids (kg)} \times \text{Dry matter content (\%)} = Q_y\ (\text{t}) \times 1000\ \text{kg} \times \left(1 - \text{Moisture content (\%)}\right)$
Comments	Three samples were extracted daily from the separated solids for testing. The moisture content was initially determined using a laboratory oven, followed by the analysis of volatile matter using a muffle furnace. The finalized daily results, which was the average value of these 3 testing results, were recorded in the measuring records and the monthly mean was calculated

The monitoring equipment employed during this monitoring period included a scale, a laboratory oven and an analytical balance. According to the "Catalog of Measuring Instruments Subject to Mandatory Management"[21] released by the State Administration for Market Regulation of China in 2019, the scale and the analytical balance are subject to compulsory calibration:

Monitoring Equipment	Scale	Analytical Balance
Type	FSG-30-QC	FA2104
Serial Number	20210818111	SHP0200578128
Accuracy	III	1mg
Calibration Date	21-Mar-2022 21-Mar-2023	21-Mar-2022 21-Mar-2023
Validity Date	20-Mar-2023 20-Mar-2024	20-Mar-2023 20-Mar-2024

Information of other monitoring equipment is outlined below:

Measuring Equipment	Laboratory Oven
Type	DHG-9050B
Serial Number	15041529
Accuracy	±1°C

Data / Parameter	$VS_{ss,y}$
Data unit	kg/kg
Description	Volatile solids content of the separated solids in year y on a dry matter basis

Value applied:

Retrieved from the measuring records of the 1st monitoring period and averaged on a calendar year basis:

Vintage	Value (Swine)
2022	0.55
2023	0.56

Comments

Three samples were extracted daily from the separated solids for testing. The moisture content was initially determined using a laboratory oven, followed by the analysis of volatile matter using a muffle furnace. The finalized daily results, which was the average

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[21] https://www.samr.gov.cn/jls/zt/jlfwzxqy/zcfg/art/2023/art_1d7270734e844327aec3ff6738dbf9b0.html

<table>
<tr><td></td><td>value of these 3 testing results, were recorded in the measuring records and the monthly mean was calculated
The monitoring equipment employed during this monitoring period included a Muffle furnace and an analytical balance. According to the “Catalog of Measuring Instruments Subject to Mandatory Management”[21] released by the State Administration for Market Regulation of China in 2019, the analytical balance is subject to compulsory calibration. The analytical balance for measuring $VS_{ss,y}$ is identical to the one employed for $M_{ss,y}$, please refer to the “Comment” section of $M_{ss,y}$ above for detailed information regarding the analytical balance
Information of other monitoring equipment is outlined below:
<table>
<tr><td>Measuring Equipment</td><td>Muffle furnace</td></tr>
<tr><td>Type</td><td>SX-4-10</td></tr>
<tr><td>Serial Number</td><td>-</td></tr>
<tr><td>Accuracy</td><td>-</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><td>Data / Parameter</td><td>Q_y</td></tr>
<tr><td>Data unit</td><td>t/yr</td></tr>
<tr><td>Description</td><td>Quantity of waste composted in year y (wet basis)</td></tr>
<tr><td>Value applied:</td><td>Retrieved from the measuring records of the 1st monitoring period and aggregated on a calendar year basis
<table>
<tr><th>Vintage</th><th>Q_y</th></tr>
<tr><td>2022 (28-Mar-2022 to 31-Dec-2022)</td><td>40,456.29</td></tr>
<tr><td>2023 (01-Jan-2023 to 30-Sep-2023)</td><td>38,099.20</td></tr>
<tr><td>Total</td><td>78,555.49</td></tr>
</table>
</td></tr>
<tr><td>Comments</td><td>The monitoring equipment employed during this monitoring period included a scale. Please refer to the “Comment” section of $M_{ss,y}$ above for detailed information regarding the scale</td></tr>
</table>

<table>
<tr><td>Data / Parameter</td><td>$EC_{PJ,compost,y}$</td></tr>
<tr><td>Data unit</td><td>MWh/yr</td></tr>
</table>

<table>
<tr><td>Description</td><td>Quantity of electricity consumed by the project electricity consumption from composting in year y</td></tr>
<tr><td>Value applied:</td><td>Retrieved from the monthly measuring records of the 1st monitoring period and aggregated on a calendar year basis:
<table>
<tr><th>Vintage</th><th>$EC_{PJ,project,y}$</th><th>$EC_{PJ,compost,y}$</th></tr>
<tr><td>2022 (28-Mar-2022 to 31-Dec-2022)</td><td>12.219</td><td>12.219</td></tr>
<tr><td>2023 (01-Jan-2023 to 30-Sep-2023)</td><td>9.573</td><td>9.573</td></tr>
<tr><td>Total</td><td>21.792</td><td>21.792</td></tr>
</table>
Values above are calculated as follows:

$$EC_{PJ,compost,y} = EC_{PJ,project,y}$$
</td></tr>
<tr><td>Comments</td><td>The metering records were crosschecked against the payment receipts

The monitoring equipment employed during this monitoring period included an electricity meter. According to the “Catalog of Measuring Instruments Subject to Mandatory Management”[22] released by the State Administration for Market Regulation of China in 2019, the electricity meter is subject to compulsory calibration:
<table>
<tr><th>Monitoring Equipment</th><th>Electricity Meter</th></tr>
<tr><td>Type</td><td>DSZ331</td></tr>
<tr><td>Serial Number</td><td>00000025117515</td></tr>
<tr><td>Accuracy</td><td>0.5</td></tr>
<tr><td>Calibration Date</td><td>24-Jan-2022
24-Jan-2023</td></tr>
<tr><td>Validity Date</td><td>23-Jan-2023
23-Jan-2024</td></tr>
</table>
</td></tr>
</table>

Data / Parameter	$N_{LT,y}$
Data unit	Number
Description	Number of swine for the year y

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[22] https://www.samr.gov.cn/jls/zt/jlfwzxqy/zcfg/art/2023/art_1d7270734e844327aec3ff6738dbf9b0.html

<table>
<tr><td>Value applied:</td><td>
Retrieved from the annual operation records of the 1st monitoring period and aggregated on a calendar year basis
<table>
<tr><th>Vintage</th><th>$N_{da,y}$</th><th>$N_{p,y}$</th><th>$N_{LT,y}$</th></tr>
<tr><td>2022 (28-Mar-2022 to 31-Dec-2022)</td><td>279</td><td>84,558</td><td>64,634</td></tr>
<tr><td>2023 (01-Jan-2023 to 30-Sep-2023)</td><td>273</td><td>80,286</td><td>60,049</td></tr>
<tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr>
</table>
Values above are calculated as follows:

$$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$$
</td></tr>
<tr><td>Comments</td><td>This parameter does not affect the ER calculations. The PP also collected and consolidated food purchasing data from swine farms to ensure validity</td></tr>
</table>

<table>
<tr><td>Data / Parameter</td><td>$VS_{LT,y}$</td></tr>
<tr><td>Data unit</td><td>kg per year of VS excreted by swine</td></tr>
<tr><td>Description</td><td>Annual amount of volatile solids excreted by swine managed by the management system in year y</td></tr>
<tr><td>Value applied:</td><td>
Retrieved from Chapter 10, Volume 4, 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
<table>
<tr><th>Vintage</th><th>Value</th></tr>
<tr><td>2022 (28-Mar-2022 to 31-Dec-2022)</td><td>108.30</td></tr>
<tr><td>2023 (01-Jan-2023 to 30-Sep-2023)</td><td>108.30</td></tr>
<tr><td>Total</td><td>-</td></tr>
</table>
Values above are calculated as follows:

$$VS_{swine,y} = 4.3\text{kg}/1000\text{kg animal mass}/\text{day} \times 69\text{kg animal mass}/\text{hd} \times 365\text{days} = 108.30\text{kg}/\text{hd}/\text{y}$$
</td></tr>
<tr><td>Comments</td><td>This parameter does not affect the ER calculations</td></tr>
</table>

<table>
<tr><td>Data / Parameter</td><td>$Q_{y,transp}$</td></tr>
</table>

Data unit	tonnes
Description	Quantity of matters transported in year y
Value applied:	Retrieved from the inventory records of the 1st monitoring period and aggregated on a calendar year basis
Comments	The monitoring equipment employed during this monitoring period included a scale. Please refer to the "Comment" section of $M_{ss,y}$ above for detailed information regarding the scale

Vintage	$Q_{y,internal}$	$Q_{y,external}$	$Q_{y,transp}$
2022 (28-Mar-2022 to 31-Dec-2022)	0	102,316	102,316
2023 (01-Jan-2023 to 30-Sep-2023)	0	86,011	86,011
Total	0	188,327	188,327

Values above are calculated as follows:

$$Q_{y,transp} = Q_{y,internal} + Q_{y,external}$$

7.2 Baseline Emissions

The ex-post calculation of the baseline emissions (BE_y) of the 1st monitoring period are calculated as per Equation (1), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$B_{o,w,y}$	0.45	0.45	$m^3CH_4/kg\text{-}VS$	Default, IPCC 2019 refinement, Table 10.16A, Swine, Other Regions, High productivity systems
$M_{ss,y}$	16,093,939	15,184,739	kg	Monitored
$VS_{ss,y}$	0.55	0.56	kg/kg	Monitored
UF_b	0.94	0.94	-	Default, AMS-III.Y
GWP_{CH4}	28.00	28.00	$tCO_2e\ /tCH_4$	Default, IPCC AR5

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[23] 28-Mar-2022 to 31-Dec-2022, a total of 279 days.

[24] 01-Jan-2023 to 30-Sep-2023, a total of 273 days.

D_{CH4}	0.67	0.67	kg/m3	Default, AMS-III.Y
$MS_{Bl,i}$	100%	100%	-	Default, AMS-III.Y. There was only one type of treatment system existed prior to the project implementation, as per the Feasibility Study Report
$MCF_{b,i}$	73%	73%	-	Default, IPCC 2019 refinement, Table 10.17, Uncovered anaerobic lagoon. The project falls within warm, temperate, moist climate zone. There was only one type of treatment system existed prior to the project implementation, as per the Feasibility Study Report
BE_y	51,276	49,259	tCO_2e	Calculated

7.3 Project Emissions

The ex-post calculation of the project emissions (PE_y) of the 1st monitoring period are calculated as per Equation (2), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$PE_{y,ss}$	5,269	4,958	tCO_2e	Calculated
$PE_{y,power}$	-	-	tCO_2e	-
$PE_{y,floc,combu}$	-	-	tCO_2e	-
$PE_{y,transp}$	2,098	1,764	tCO_2e	Calculated
PE_y	7,367	6,722	tCO_2e	Calculated

1) The ex-post calculation of the project emissions from storage, use, destruction or disposal of solids separated ($PE_{y,ss}$) of the 1st monitoring period are calculated as per Equation (3) and Equation (4), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$\mathrm{PE}_{\mathrm{EC,y}}$	20	15	tCO_2e	Calculated
$\mathrm{PE}_{\mathrm{FC,y}}$	838	789	tCO_2e	Calculated
$\mathrm{PE}_{\mathrm{CH4,y}}$	2,266	2,134	tCO_2e	Calculated
$\mathrm{PE}_{\mathrm{N2O,y}}$	2,145	2,020	tCO_2e	Calculated
$\mathrm{PE}_{\mathrm{RO,y}}$	-	-	tCO_2e	-
$\mathrm{PE}_{\mathrm{y,ss}}$ or $\mathrm{PE}_{\mathrm{s,composting}}$	5,269	4,958	tCO_2e	Calculated

a) The ex-post calculation of the composting emissions from electricity consumption ($\mathrm{PE}_{\mathrm{EC,y}}$) of the 1st monitoring period are calculated as per Equation (5) and Equation (6), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$\mathrm{EC}_{\mathrm{PJ,compost,y}}$ or $\mathrm{EC}_{\mathrm{PJ,project,y}}$	12.219	9.573	MWh/yr	Feasibility Study Report
$\mathrm{EF}_{\mathrm{EF,compost,y}}$	1.3	1.3	tCO_2e/MWh	Default, Tool 5
$\mathrm{TDL}_{\mathrm{compost,y}}$	0.2	0.2	-	Default, Tool 5
$\mathrm{PE}_{\mathrm{EC,y}}$	20	15	tCO_2e/yr	Calculated

b) The ex-post calculation of the composting emissions from fossil fuel consumption ($\mathrm{PE}_{\mathrm{FC,y}}$) of the 1st monitoring period are calculated as per Equation (7), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$\mathrm{EF}_{\mathrm{FC,default}}$	0.0207	0.0207	tCO_2/t	Default, Tool 13
Q_{y}	40,456.29	38,099.20	t/yr	Monitored
$\mathrm{PE}_{\mathrm{FC,y}}$	838	789	tCO_2e/yr	Calculated

c) The ex-post calculation of the composting emissions of methane ($\mathrm{PE}_{\mathrm{CH4,y}}$) are calculated as per Equation (8), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$\mathrm{EF}_{\mathrm{CH4,y}}$	0.002	0.002	tCH_4/t	Default, Tool 13

GWP_{CH4}	28	28	tCO_2e/tCH_4	Default, IPCC AR5
Q_y	40,456.29	38,099.20	t/yr	Monitored
$PE_{CH4,y}$	2,266	2,134	tCO_2e/yr	Calculated

d) The ex-post calculation of the composting emissions of nitrous oxide ($PE_{N2O,y}$) are calculated as per Equation (9), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$EF_{N2O,y}$	0.0002	0.0002	tN_2O/t	Default, Tool 13
GWP_{N2O}	265	265	tCO_2e/tN_2O	Default, IPCC AR5
Q_y	40,456.29	38,099.20	t/yr	Monitored
$PE_{N2O,y}$	2,145	2,020	tCO_2e/yr	Calculated

e) The ex-post calculation of the projection emissions from incremental transportation of matters ($PE_{y,transp}$) are calculated as per Equation (12), incorporating the following specified parameters:

Parameter	2022[23]	2023[24]	Unit	Sources
$Q_{y,transp}$	102,316	86,011	t	Monitored
CT_y	8	8	t/truck	Monitored
DT_y	200	200	km/truck	Monitored
EF_{CO2}	0.00082	0.00082	tCO_2e/km	Default, Local values & IPCC
$PE_{y,transp}$	2,098	1,764	tCO_2e	Calculated

7.4 Leakage

During the 1st monitoring period, the project did not have any leakage emissions to be addressed, due to the following reasons:

1) Solid separation technologies of the project were brand new at the time of commissioning and were not transferred from another activity;

2) The project did not use flocculants during this monitoring period.

7.5 Net GHG Emission Reductions and Removals

The ex-post calculation of the emission reductions ($\mathrm{ER_y}$) of the 1st monitoring period are calculated as per Equation (13). The results are shown in the table below. Since the monitoring period begins and ends part-way through 2022 and 2023, these years' total reductions are adjusted (10 months of credit in 2022, 9 months of credit in 2023). The total emission reduction during this monitoring period of 28-Mar-2022 to 30-Sep-2023 is 88,210tCO$_2$e.

Year	Baseline emissions or removals (tCO$_2$e)	Project emissions or removals (tCO$_2$e)	Leakage emissions (tCO$_2$e)	Net GHG emission reductions or removals (tCO$_2$e)
2022 (28-Mar-2022 to 31-Dec-2022)	51,276	7,367	0	43,909
2023 (01-Jan-2023 to 30-Sep-2023)	49,259	4,958	0	44,301
Total	**100,535**	**12,325**	**0**	**88,210**

As per Section 5.4, the expected annual emission reduction is 57,307tCO$_2$e, so the expected emission reduction during the 1st monitoring period from 28-Mar-2022 to 30-Sept-2023 (552 days) is 86,667tCO$_2$e[25]. The achieved emission reductions of the 1st monitoring period are slightly above the ex-ante expected level by 1.78%[26], which is considered within the normal range of variation.

Ex-ante emissions reductions/ removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
86,667	88,210	1.78%	Variation deemed within the normal range.

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[25] $57{,}307\mathrm{tCO_2e} \div 365\ \mathrm{days} \times (552\ \mathrm{days}) = 86{,}667\mathrm{tCO_2e}$

[26] $(88{,}210\mathrm{tCO_2e} \div 86{,}667\mathrm{tCO_2e} - 1) \times 100\% = 1.78\%$

APPENDIX 1: SUPPORTING EVIDENCE

Evidence for SDG 8

Employment statistics derived from the human resource database of this project have been submitted to VVB for verification.